Highlights

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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- Closed-loop knowledge-guided swarm learning for high-D feature selection.
- MI seeding with adaptive recovery prevents stagnation and restores diversity.
- Knowledge-guided elite refinement accumulates feature signals with decay.
- Two-phase weight scheduling decouples feature pruning from accuracy tuning.
- KDS-EL achieves 95.74% accuracy with only 87.8 features on 20 benchmarks.

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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ABSTRACT

Feature selection in high-dimensional data remains a central challenge in evolutionary machine learning. Particle swarm optimization suffers from three structural deficiencies: undirected initialization provides insufficient search guidance in sparse high-dimensional spaces, uniform feature perturbation fails to exploit cross-iteration feature-importance feedback, and static objective weighting cannot adapt to shifting evolutionary demands. To overcome these bottlenecks, this paper proposes Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL), a closed-loop framework incorporating domain prior injection, generational knowledge accumulation, and dynamic objective scheduling. Specifically, mutual-information-guided seeding provides continuous relevance priors while adaptive re-initialization restores population diversity upon stagnation. A knowledge-guided elite refinement operator dynamically harvests cross-iteration feature feedback with exponential memory decay, steering elite solutions toward historically effective dimensions under bounded computational overhead. Furthermore, two-phase weight scheduling temporally decouples early coarse dimensionality reduction from late-stage accuracy refinement. Extensive evaluations across 20 ultra-high-dimensional benchmark datasets demonstrate that KDS-EL achieves a leading average classification accuracy of 95.74% while isolating an ultra-sparse signature of only 87.8 features, significantly outperforming 11 representative competitor algorithms and foundational multi-task frameworks. Non-parametric statistical tests with Holm–Bonferroni correction confirm the statistically significant superiority and computational efficiency of the proposed framework.

1. Introduction

High-dimensional data have emerged as a fundamental challenge in classification problems across diverse domains, including genomics, text mining, and computer vision. These datasets typically contain hundreds or even tens of thousands of features, while only a limited number of labeled samples are available, resulting in the well-known $p \gg n$ scenario. Under such conditions, conventional learning algorithms often suffer from severe overfitting, increased computational complexity, and reduced interpretability due to the curse of dimensionality [1, 2]. Feature selection (FS) has therefore become an essential preprocessing technique that aims to identify a compact and informative subset of features while preserving classification performance. However, selecting an optimal feature subset remains highly challenging because the search space grows exponentially with the number of features D, and the underlying subset-selection problem is NP-hard [3]. Moreover, individual feature relevance measures are insufficient to characterize complex feature interactions and dependencies, which are often critical for achieving robust classification performance.

FS methods are generally categorized into filter, wrapper, and embedded approaches [1, 4]. Filter methods rank features according to statistical criteria and offer high computational efficiency, but they usually evaluate features independently of the classifier and may overlook feature interactions relevant to classification. Embedded methods incorporate feature selection into the learning process, providing a compromise between selection efficiency and predictive performance; however, their effectiveness is often dependent on the characteristics of specific learning models. Wrapper methods directly optimize feature subsets according to classifier performance and can achieve competitive classification results, but the repeated model evaluations make them computationally expensive, especially in high-dimensional search spaces. To address the combinatorial complexity of wrapper-based FS, evolutionary computation has been extensively investigated as a promising optimization paradigm [3, 5–7]. By maintaining a population of candidate solutions, evolutionary algorithms can perform global exploration of feature subsets without requiring gradient information. Among them, particle swarm optimization (PSO), genetic algorithms, and differential evolution have demonstrated promising results on high-dimensional FS tasks.

PSO has become one of the most widely adopted evolutionary algorithms for FS due to its simple structure, efficient information-sharing mechanism, and fast convergence. Nevertheless, conventional PSO-based FS methods still encounter three structural limitations when dealing with sparse and high-dimensional search spaces. First, random initialization distributes particles without prior guidance throughout the binary search space, whereas effective feature subsets usually occupy only a small region with limited active dimensions. Consequently, a considerable number of fitness evaluations may be wasted on redundant and suboptimal solutions before the swarm identifies promising search regions. Second, conventional uniform feature perturbation strategies treat all dimensions equally and neglect the feature-level improvement information accumulated during previous evaluations, preventing the search from exploiting historically informative features. Third, existing PSO-based FS methods typically employ fixed objective weights to balance classification error and feature subset size during the entire optimization process. However, the desired trade-off between feature reduction and accuracy refinement varies across search stages: aggressive sparsity enhancement is preferred in early exploration, whereas accurate decision boundary refinement becomes more important after a compact subset has been obtained.

Recent evolutionary FS studies have attempted to improve search performance through enhanced population initialization, adaptive evolutionary operators, and refined fitness evaluation strategies [8–11]. Although these approaches have achieved promising results, they usually focus on isolated components of the optimization process and do not jointly address inefficient initialization, insufficient exploitation of accumulated feature knowledge, and inflexible objective weighting. Recently, multi-task PSO has emerged as an effective paradigm for high-dimensional FS with fixed-length binary representations. For instance, MEL [12] adopts cooperative subpopulations and a fixed-length representation to efficiently handle feature spaces containing tens of thousands of dimensions. Despite its effectiveness, existing multi-task PSO-based FS frameworks still leave several challenges unresolved: the initial population is generally generated without sufficient feature-level prior knowledge, useful feature-importance information discovered during evolution is not explicitly retained and exploited, and the balance between classification performance and feature sparsity remains unchanged across different search stages.

To address these limitations, this paper proposes Knowledge-Guided Dual-Phase Swarm Evolutionary Learning (KDS-EL), a closed-loop framework that cohesively integrates domain prior injection, cross-iteration knowledge accumulation, and dynamic objective scheduling into a cooperative multi-task PSO backbone.

The main contributions of this paper are summarized as follows:

1. *MI-Guided Initialization with Adaptive Re-initialization*: A hybrid seeding strategy that injects continuous feature-relevance priors into population construction to eliminate early-stage search waste, coupled with an adaptive re-initialization mechanism that restores swarm diversity upon evolutionary stagnation (Section 3.2).
2. *Knowledge-Guided Elite Refinement (KER)*: A targeted elite perturbation operator that dynamically harvests cross-iteration feature-importance feedback with exponential memory decay, directing elite solutions toward historically effective features within a strictly bounded evaluation overhead (Section 3.3).

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Table 1: Methodological taxonomy of investigated evolutionary feature selection algorithms.

		Knowledge Engineering			Search Control		
Algorithm	Paradigm	MI Prior Seeding	Online Memory	Memory Decay	Elite Refinement	Dynamic Weighting	Complexity
BPSO [7]	SO-SI	×	×	×	×	×	High
FWPSO [8]	SO-SI	○	×	×	×	×	Moderate
SaWDE [9]	SO-DE	×	×	×	×	○	High
IGPSO [10]	SO-SI	○	×	×	×	×	Moderate
PSO-CSM [11]	SO-SI	○	×	×	×	×	Moderate
DENCA-SO [14]	SO-DE	○	×	×	×	×	Low
bIRIME [13]	SO-SI	×	×	×	○	×	High
BinCOA [15]	SO-SI	×	×	×	○	×	High
MEL [12]	MT-SI	×	✓	×	×	×	High
KDS-EL (Proposed)	**MT-SI**	✓	✓	✓	✓	✓	**Low**

✓: Fully supported; ○: Partially supported; ×: Absent / not supported.
Paradigms: SO: Single-Objective; MT: Multi-Task; SI: Swarm Intelligence; DE: Differential Evolution.

3. *Two-Phase Weight Scheduling (TPS)*: A stage-aware objective scheduling mechanism that temporally decouples early aggressive feature pruning from late-stage accuracy refinement, overcoming the rigidity of static weighting (Section 3.4).
4. *Comprehensive Empirical Validation*: Extensive evaluations across 20 high-dimensional benchmark datasets (spanning 500 to 54,675 features) against state-of-the-art single-objective competitors and precursor frameworks, rigorously validated through non-parametric statistical testing, component contribution analysis, and parametric sensitivity investigations (Section 4).

Extensive experiments demonstrate that KDS-EL achieves the highest classification performance, obtaining an average classification accuracy of 0.9574 with only 87.8 selected features, corresponding to an average feature selection ratio of merely 1.1%. Compared with recent top-tier competitor bIRIME [13] and established competitor FWPSO [8], KDS-EL improves classification accuracy by 1.20 and 3.97 percentage points, while reducing the selected feature count by 78.5% and 91.8%, respectively. Furthermore, KDS-EL achieves a 3.4× speedup over the foundational multi-task precursor MEL [12], reducing average runtime from 175.0 s to 51.3 s. Pairwise Wilcoxon signed-rank tests with Holm–Bonferroni step-down correction confirm that the performance advantages are statistically significant over all 11 comparison algorithms and ablative variants.

The remainder of this paper is organized as follows. Section 2 formulates the feature selection optimization problem and reviews related literature and background concepts. Section 3 presents the proposed KDS-EL framework and formulates its three constituent mechanisms in detail. Section 4 presents comprehensive experimental results, statistical tests, component contribution analysis, and parameter sensitivity investigations. Finally, Section 5 concludes the paper and discusses prospective research directions.

2. Related Work and Background

This section establishes the formal mathematical foundation of feature selection and systematically reviews the relevant evolutionary computation paradigms. Section 2.1 first formalizes the combinatorial optimization problem and categorizes classical search paradigms. Section 2.2 subsequently analyzes evolutionary swarm mechanics alongside the structural failure modes of classical stochastic binarization in high-dimensional spaces. To address high-dimensional search challenges, Section 2.3 scrutinizes the cooperative multi-task evolutionary learning framework and pinpoints its operational bottlenecks. Finally, Section 2.4 investigates knowledge-guided evolutionary search, contrasts offline filter priors against online closed-loop memory, and establishes the taxonomic positioning of KDS-EL in contrast to existing literature.

2.1. Problem formulation and search paradigms in feature selection

The essence of feature selection lies in combinatorial optimization, aiming to identify an optimal, minimal subset of informative features from a high-dimensional feature space to maximize classification performance while eliminating redundant and uninformative dimensions [1, 2]. Formally, let $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ denote a supervised training dataset comprising N_{samp} instances, where $\mathbf{f}_i \in \mathbb{R}^D$ represents the full feature vector in the original D-dimensional space and $y_i \in \{1, 2, \ldots, C\}$ designates the discrete class label across C categories. In discrete binary representations, a candidate feature subset is formulated as a binary decision indicator vector $\mathbf{x} \in \{0, 1\}^D$, defined as:

$$x_j = \begin{cases} 1, & \text{if feature } f_j \text{ is selected,} \\ 0, & \text{if feature } f_j \text{ is discarded.} \end{cases} \tag{1}$$

The primary objective of feature selection is to simultaneously minimize the classification error rate and the cardinality of the selected feature subset. In evolutionary wrapper feature selection, this multi-criterion goal is mathematically formalized as minimizing a scalarized single-objective fitness function $f(\mathbf{x})$ [16, 17]:

$$\min_{\mathbf{x}\in\{0,1\}^D} f(\mathbf{x}) = w_a \cdot \text{ErrorRate}(\mathbf{x}) + w_f \cdot \left(\frac{\|\mathbf{x}\|_0}{D}\right), \tag{2}$$

subject to $\mathbf{x} \in \{0, 1\}^D$ and $w_a + w_f = 1$, where ErrorRate($\mathbf{x}$) represents the cross-validation classification error evaluated on the masked feature subset $\{\mathbf{f}_i \odot \mathbf{x}\}$ using an inductive k-nearest neighbors (k-NN) learner, with $\odot$ denoting the Hadamard element-wise product. The term $\|\mathbf{x}\|_0$ denotes the number of active features, $\|\mathbf{x}\|_0/D$ represents the feature selection ratio, and $w_a, w_f \in [0, 1]$ serve as trade-off coefficients balancing predictive accuracy against subset compactness. Because the cardinality of the discrete search space is 2^D, evaluating all candidate subsets exhaustively is NP-hard and computationally intractable when $D \gg 10^3$. Traditional feature selection techniques are broadly categorized into three classical paradigms based on their evaluation criteria: filter, embedded, and wrapper methods [2, 4]. Specifically, *filter methods* evaluate feature subsets based solely on intrinsic statistical or information-theoretic metrics—such as mutual information in mRMR [18], correlation in FCBF [19], or centrality scores [20]—independently of any classification algorithm. While filter methods are computationally lightweight with $O(DN_{\text{samp}} + D \log D)$ complexity, their classifier-agnostic evaluation and rigid hard truncation permanently discard weakly relevant features that exhibit strong collective synergy when combined. Conversely, *embedded methods* perform feature selection natively during classifier training by embedding sparsity constraints into model parameter estimation, such as L_1-regularized regression in Bolasso [21] or tree-based ensemble importance in FS-GBDT [22]. Although embedded methods bridge computational efficiency and predictive feedback, they remain architecturally bound to specific learners and frequently encounter performance degradation under severe sample scarcity where $N_{\text{samp}} \ll 100$. In contrast, *wrapper methods* utilize a target learning algorithm as a black-box evaluator, assessing candidate feature masks directly through cross-validation predictive error feedback [4, 16]. By tightly coupling subset evaluation with classifier feedback, wrappers naturally capture complex non-linear feature interactions and yield superior classification accuracy.

To navigate the discrete 2^D combinatorial space in wrapper feature selection without gradient information, Evolutionary Computation paradigms—including Genetic Algorithms [23], Ant Colony Optimization [24], Artificial Bee Colony [25], and Swarm Intelligence [3, 26]—have emerged as dominant search heuristics. To combat the curse of dimensionality in small-sample high-dimensional data, various advanced single-objective evolutionary strategies have been introduced. For instance, self-adaptive differential evolution with adaptive weighting (SaWDE) [9] dynamically

balances classification error against subset cardinality; feature-weight particle swarm optimization (FWPSO) [8] injects offline feature weights to steer velocity updates; importance-guided particle swarm optimization (IGPSO) [10] employs multi-layer perceptron importance scores to bias particle movements; and particle swarm optimization with comprehensive scoring mechanism (PSO-CSM) [11] leverages multi-criterion feature evaluation to restrict search trajectories. Despite these advancements, standard single-objective evolutionary wrappers face severe structural bottlenecks in ultra-high-dimensional spaces where $D \gg 10^3$. Uniform random initialization scatters individuals indiscriminately across the search space, generating overly dense candidate masks with approximately $D/2$ active features that burden classifier evaluations and provoke premature overfitting. Furthermore, stochastic variation operators treat feature perturbations as memoryless trials, discarding historical feature-importance feedback across generations. Finally, static or coarsely tuned objective weighting enforces rigid trade-offs throughout evolution, failing to adapt to the distinct requirements of early-stage dimensional reduction and late-stage accuracy refinement.

2.2. Evolutionary swarm intelligence and discrete feature selection dynamics

Standard particle swarm optimization, originally introduced by Kennedy and Eberhart [5], is a population-based stochastic meta-heuristic inspired by the collective foraging behaviors of bird flocks and fish schools. A swarm consists of N candidate particles navigating a D-dimensional continuous solution space $\mathbb{R}^D$. At iteration t, each particle i is characterized by a continuous position vector $\mathbf{p}_i^{(t)} \in \mathbb{R}^D$ and an associated velocity vector $\mathbf{v}_i^{(t)} \in \mathbb{R}^D$, steered dynamically by its personal historical best position $\mathbf{p}_{\text{best},i} \in \mathbb{R}^D$ and the swarm's global best position $\mathbf{g}_{\text{best}} \in \mathbb{R}^D$. The velocity and position updates of particle i at iteration $t+1$ are formulated as follows [6]:

$$\mathbf{v}_i^{(t+1)} = \omega \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},i} - \mathbf{p}_i^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best}} - \mathbf{p}_i^{(t)}), \tag{3}$$

$$\mathbf{p}_i^{(t+1)} = \mathbf{p}_i^{(t)} + \mathbf{v}_i^{(t+1)}, \tag{4}$$

where $\omega \in [0, 1]$ denotes the inertia weight controlling momentum retention, $c_1, c_2 > 0$ are the cognitive and social acceleration coefficients governing the attraction toward individual and collective best experiences, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are independent uniform random factors providing stochastic exploration capability.

To adapt continuous PSO to the combinatorial nature of binary feature selection where candidate solutions are represented as binary indicator masks $\mathbf{x}_i \in \{0, 1\}^D$, Kennedy and Eberhart [7] proposed Binary PSO (BPSO). In BPSO, continuous velocities are mapped into selection probabilities via a Sigmoid transfer function:

$$S(v_{i,j}^{(t+1)}) = \frac{1}{1 + \exp(-v_{i,j}^{(t+1)})}, \tag{5}$$

and the corresponding binary decision variable $x_{i,j}^{(t+1)}$ is sampled probabilistically as:

$$x_{i,j}^{(t+1)} = \begin{cases} 1, & \text{if } r_{i,j} < S(v_{i,j}^{(t+1)}), \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

where $r_{i,j} \sim \mathcal{U}(0, 1)$ is an independent uniform random threshold. Although BPSO exhibits fast convergence on low-dimensional benchmarks, it encounters severe exploration-exploitation dilemmas when applied to ultra-high-dimensional biomedical domains where $D \gg 10^3$. Because the Sigmoid function satisfies $S(0) = 0.5$, continuous velocities hovering near zero inevitably yield an expected active feature density of approximately $D/2$. In typical biomedical datasets containing $D \geq 10{,}000$ dimensions, this stochastic mapping generates candidate subsets with more than 5,000 active features, heavily overloading wrapper classifier evaluations with $O(N_{\text{samp}}^2 d_{\text{selected}})$ complexity and triggering severe overfitting under small sample sizes where $N_{\text{samp}} \ll 100$. Furthermore, continuous stochastic sampling introduces high generational variance: even when velocity vectors stabilize, candidate feature masks continue to jitter stochastically, causing particle trajectories to degrade into memoryless random walks across the combinatorial space.

To overcome the random drift of standard binarization in high-dimensional feature selection, diverse single-objective meta-heuristic algorithms have recently been proposed. Evolutionary filter methods, such as DENCA-SO [14], employ neighborhood component feature selection (NCFS) continuous weights within differential evolution operators. By evaluating candidate subsets using algebraic surrogate costs rather than wrapper classifiers, DENCA-SO achieves rapid wall-clock execution; however, because its optimization process remains entirely decoupled from inductive classification margins, it suffers from severe feature redundancy and retains over 48% of original dimensions on biomedical benchmarks. Concurrently, advanced binary swarm optimizers seek to enhance discrete exploration. For instance, bIRIME [13] incorporates dynamic escape energy, composite mutation strategies, and opposition-based restart mechanisms to mitigate premature convergence, whereas BinCOA [15] couples refractive opposition-based learning with crossed horizontal and vertical search operations. Nevertheless, these multi-operator strategies incur prohibitive computational burdens: generating and evaluating multiple trial vectors per agent scales wrapper classifier calls to three or four times the population size per generation, causing execution times to escalate beyond 2,000 seconds on large-sample benchmarks (Section 4.2.6). Crucially, in ultra-high dimensions without proactive pruning guidance, stochastic binarization traps these algorithms near the uninformative 50% dimensionality attractor, preserving thousands of redundant features. On the other hand, wrapper methods employing scalarized fitness formulations—such as SaWDE [9], FWPSO [8], and PSO-CSM [11]—enforce fixed static objective weights throughout evolution. This static weighting creates a fundamental evolutionary phase misalignment: the search process inherently requires aggressive feature pruning during early exploration to eliminate vast noise subspaces, whereas delicate decision-boundary fine-tuning is required during late-stage exploitation to maximize predictive accuracy.

2.3. Multi-task evolutionary learning and the MEL framework

Evolutionary Multi-Task Optimization (EMTO) aims to solve multiple distinct optimization tasks simultaneously by facilitating inter-task knowledge transfer through a unified chromosomal representation [27]. In feature selection, EMTO frameworks partition the global population into collaborative subpopulations to explore complementary search subspaces, including minion-assisted multi-tasking [28], evolutionary multi-task PSO for hyperspectral and gene expression data [29, 30], multi-task bee colony band selection [31], and large-scale evolutionary multi-task optimization [32]. Most notably, Wang et al. [12] introduced Multi-Task Evolutionary Learning (MEL), which establishes a dual-subpopulation PSO framework. In MEL, a population of N particles is partitioned into two collaborative subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$, with each subpopulation comprising $N/2$ individuals operating on complementary search tasks. Specifically, *Subpopulation 1* executes continuous swarm exploration, updating continuous velocities and positions using PSO dynamics augmented with an inter-task knowledge transfer channel directing particles toward the historical global best of Subpopulation 2 ($\mathbf{g}_{\text{best},2} \in \mathbb{R}^D$):

$$\mathbf{v}_{1,i}^{(t+1)} = \omega \mathbf{v}_{1,i}^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},1,i} - \mathbf{p}_{1,i}^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best},1} - \mathbf{p}_{1,i}^{(t)}) + c_3 r_3 (\mathbf{g}_{\text{best},2} - \mathbf{p}_{1,i}^{(t)}), \tag{7}$$

where $\omega \in [0, 1]$ is the inertia weight, $c_1, c_2, c_3 > 0$ are acceleration coefficients governing self-cognition, social learning, and cross-task transfer, and $r_1, r_2, r_3 \sim \mathcal{U}(0, 1)$ are independent stochastic factors. Concurrently, *Subpopulation 2* performs importance-guided discrete sampling directly from a shared feature-importance matrix $\mathbf{W}_{\text{MEL}} \in \mathbb{R}^D$, whose elements accumulate the historical selection frequencies of features across successful candidate solutions.

Both subpopulations evaluate candidate subsets using a shared scalar fitness function similar to Eq. (2) that combines classification error and feature subset ratio via fixed static trade-off weights $\alpha, \beta \in [0, 1]$ satisfying $\alpha + \beta = 1$.

While MEL demonstrates the utility of cross-task knowledge transfer, its search dynamics in ultra-high-dimensional spaces where $D \gg 10^3$ suffer from critical structural bottlenecks. Continuous particles in $\mathcal{P}_1$ are initialized uniformly across the unit hypercube, lacking domain-informed guidance toward sparse biomarker subspaces and wasting substantial evaluation budget exploring irrelevant dimensions in early iterations. Moreover, the importance matrix $\mathbf{W}_{\text{MEL}}$ accumulates historical feature frequencies unconditionally without exponential forgetting or elite fitness verification, allowing obsolete early choices to permanently bias subsequent discrete sampling in $\mathcal{P}_2$ and trapping the swarm in local optima. In addition, rigid static weighting prevents the algorithm from dynamically adjusting its selection pressure between early aggressive feature pruning and late high-precision boundary refinement. The proposed KDS-EL framework directly adopts this dual-subpopulation multi-task foundation while resolving each of these structural deficiencies through targeted mechanisms.

2.4. Knowledge-guided search and taxonomic positioning

Integrating domain-specific prior knowledge into meta-heuristics has attracted surging attention as an effective means to escape the blindness of black-box search [17, 33, 34]. In high-dimensional feature selection, Mutual Information (MI), rooted in Shannon's information theory [35], serves as the primary information-theoretic metric to quantify non-linear feature relevance. For continuous feature variables $f_j \in \mathbb{R}$ and discrete class labels $Y \in \{1, \ldots, C\}$, the

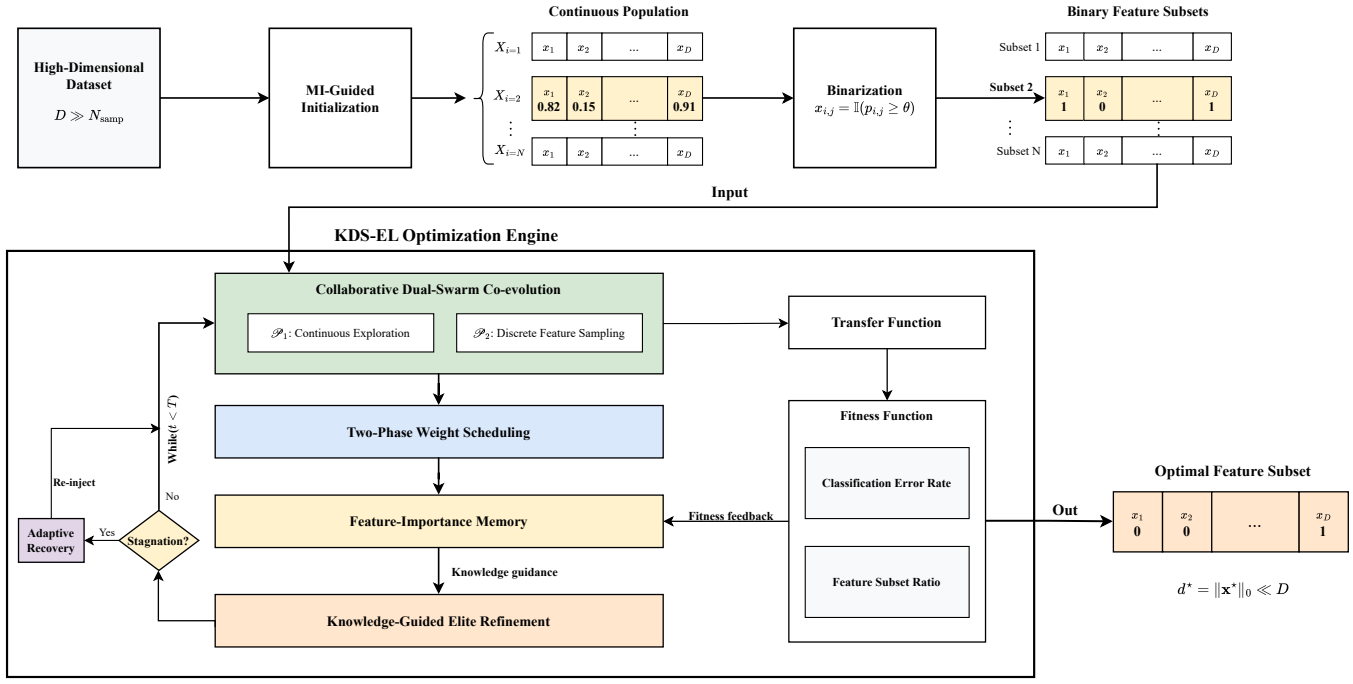


Figure 1: Example of the feature selection process via KDS-EL.

mutual information $I(f_j; Y)$ is formulated as:

$$I(f_j; Y) = H(f_j) - H(f_j \mid Y) = \sum_{c=1}^{C} P(Y=c) \int_{\mathbb{R}} p(f_j \mid Y=c) \log_2 \frac{p(f_j \mid Y=c)}{p(f_j)} \, df_j, \tag{8}$$

where $p(f_j)$ and $p(f_j \mid Y = c)$ denote the marginal and class-conditional probability density functions, respectively. In empirical applications where true underlying distributions are unknown, calculating continuous mutual information through naive data binning introduces severe bin-width sensitivity and estimation bias. Instead, non-parametric entropy estimation based on k-nearest neighbors distance statistics—specifically the Kraskov–Stögbauer–Grassberger (KSG) estimator [36] extended to mixed continuous-discrete systems by Ross [37]—directly evaluates nonlinear dependency without requiring parametric assumptions. Conventional filter feature selection methods [17, 18] evaluate mutual information across all D dimensions and apply a rigid hard truncation that retains only the top-k ranked features, permanently discarding the remainder. Despite its computational speed of $O(DN_{\text{samp}} + D \log D)$, hard truncation fundamentally ignores multi-feature synergy: individual features with moderate scores that are collectively decisive when combined are permanently eliminated from the search space.

Existing knowledge-guided evolutionary algorithms typically adopt one-shot offline injection via either static filter pre-filtering or static velocity biasing. In static filter pre-filtering, exemplified by MI-MPODE [34] and NNS-DFPG-CSA [17], algorithms compute feature rankings offline and initialize the population or restrict the search space to the top-k ranked features. In static velocity biasing, such as FWPSO [8], static feature-weight vectors are incorporated to bias particle velocity equations throughout evolution. While beneficial, one-shot knowledge injection lacks online adaptability: once injected, the static prior is never updated or refined by the empirical trial-and-error feedback generated during subsequent wrapper evaluations. Conversely, learning-based surrogates, such as neural networks [10] or Gaussian processes [38], introduce severe surrogate training overhead and risk model misspecification under sample-scarce conditions where $N_{\text{samp}} \ll 100$.

To overcome the collective limitations identified in the existing literature, KDS-EL establishes a closed-loop, knowledge-guided evolutionary learning framework that unifies offline domain priors, online memory harvesting, and stage-aware objective scheduling. Specifically, normalized mutual information scores are embedded as continuous Gaussian-perturbed seeds, preserving the full search space while guiding the swarm toward an ultra-sparse initial basin, complemented by an adaptive re-initialization mechanism to recover exploratory diversity upon stagnation (Section 3.2). Throughout evolution, cross-iteration fitness feedback is harvested into a cumulative importance matrix $\mathbf{W} \in \mathbb{R}^D$ governed by exponential forgetting, enabling the KER operator to execute targeted, directional elite flips within a strictly bounded computational budget (Section 3.3). Finally, a dynamic Two-Phase Weight Scheduling strategy temporally decouples early coarse feature pruning from late high-precision accuracy refinement, achieving simultaneous gains in classification accuracy, feature compactness, and runtime efficiency (Section 3.4). Table 1 systematically summarizes the methodological taxonomy and structural comparison between KDS-EL and all nine competitor feature selection algorithms investigated in this study across seven orthogonal technical dimensions.

3. The Proposed Method

To address the severe challenges of ultra-high-dimensional feature selection ($D \gg 10^3, N_{\text{samp}} \ll 100$), this section presents the proposed KDS-EL framework in detail. Specifically, the macro data-flow architecture and algorithmic flowchart of KDS-EL are outlined in Section 3.1. The prior-guided sparse initialization and adaptive recovery strategies are detailed in Section 3.2. Section 3.3 analyzes the search failure of conventional random perturbations, and formulates the closed-loop mechanism of cumulative importance learning and knowledge-guided elite refinement. The dynamic two-phase weight scheduling strategy is presented in Section 3.4. Finally, the overall algorithmic procedure and its computational complexity analysis are synthesized in Section 3.5.

3.1. The framework of KDS-EL

The macro data-flow architecture and problem mapping of KDS-EL for high-dimensional feature selection are systematically illustrated in Fig. 1. Given a high-dimensional dataset $\mathcal{D}$, the search pipeline begins by extracting filter-level mutual information to construct a continuous prior relevance landscape. An initial swarm of continuous positions is generated via hybrid prior seeding and deterministically mapped into sparse binary candidate masks. Throughout optimization, these candidate subsets are transferred to the KDS-EL optimization engine, which coordinates collaborative dual-swarm co-evolution, dynamic two-phase fitness evaluations, cumulative feature-importance harvesting, and targeted elite refinement in closed-loop synergy with a wrapper classification evaluation engine. Upon meeting the convergence criterion, the framework outputs the optimal compact biomarker subset $\mathbf{x}^\star$ alongside its validated classification performance.

The execution logic and operational flowchart across evolutionary generations are depicted in Fig. 2. In KDS-EL, a swarm of N candidate solutions across D features is formally represented as an $N \times D$ binary decision matrix $\mathbf{X}$:

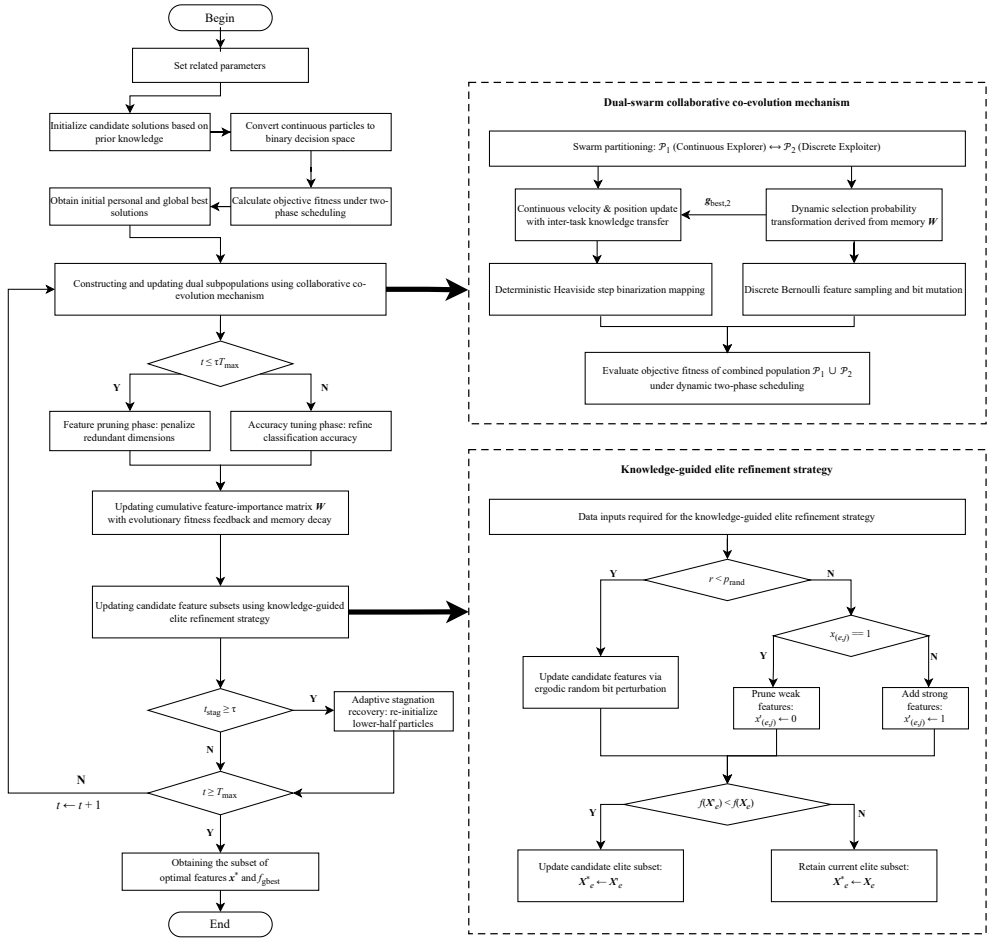


Figure 2: Overall flowchart for feature selection via KDS-EL.

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N \end{bmatrix} = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,D} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \cdots & x_{N,D} \end{bmatrix} \in \{0,1\}^{N \times D}, \tag{9}$$

where the i-th row vector $\mathbf{x}_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,D}] \in \{0,1\}^D$ denotes an individual candidate solution, with $x_{i,j} = 1$ indicating that feature f_j is selected and $x_{i,j} = 0$ indicating exclusion.

To balance global exploration and local exploitation in sparse high-dimensional spaces, the population matrix $\mathbf{X}$ is partitioned into two collaborative subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$ of equal size. Subpopulation $\mathcal{P}_1$ explores continuous coordinates via velocity–position updates with an inter-task knowledge transfer channel guided by the global best solution of $\mathcal{P}_2$. Concurrently, $\mathcal{P}_2$ conducts discrete probability-driven feature sampling informed by the accumulated feature-importance landscape. Compared with conventional single-population architectures, this dual-subpopulation design establishes complementary search dynamics: continuous swarm movement preserves momentum to explore unvisited spaces, while discrete sampling focuses evaluations on verified high-relevance dimensions, effectively mitigating premature stagnation.

During iterative optimization, KDS-EL coordinates three complementary mechanisms into a closed loop. First, population initialization injects Gaussian-perturbed mutual information priors to generate sparse candidate masks, while an adaptive re-initialization mechanism recovers population diversity whenever search progress halts. Second, differential fitness improvements across generations are accumulated into a continuous importance matrix under exponential memory decay, which actively guides the KER operator to perform targeted directional bit flips under a greedy one-to-one in-place replacement rule. Third, the dynamic TPS strategy temporally shifts objective weights across evolutionary stages, enforcing aggressive feature pruning in the early phase and fine-grained accuracy tuning in the late phase. By continuously cycling through these coordinated stages, KDS-EL systematically prunes non-informative dimensions while progressively refining compact biomarker signatures.

3.2. Population initialization and adaptive recovery

In ultra-high-dimensional feature selection, standard uniform random initialization activates each feature with equal probability, generating excessively dense candidate subsets with approximately half the features active ($d \approx D/2$). Evaluating these dense candidate subsets exhausts computational budgets on non-informative dimensions and traps search trajectories in noisy local basins. To establish an informative yet diverse initial population while mitigating mid-evolution search stagnation, KDS-EL introduces a filter-wrapper hybrid seeding strategy coupled with an adaptive stagnation recovery mechanism, as illustrated in Fig. 3.

3.2.1. Prior-guided sparse population initialization

Given the training dataset instance $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ with continuous feature vectors $\mathbf{f}_i = [f_{i,1}, \ldots, f_{i,D}] \in \mathbb{R}^D$ and discrete class labels $y_i \in \{1, \ldots, C\}$, the non-linear dependencies between individual candidate dimensions and target categories are quantified through continuous-discrete mutual information [37]. To handle continuous real-valued inputs without relying on arbitrary data discretization, the mutual information score $\hat{I}(f_j; Y)$ for each feature $j \in \{1, \ldots, D\}$ is computed via the non-parametric k-nearest neighbors KSG estimator [36, 37]:

$$\hat{I}(f_j; Y) = \psi(N_{\text{samp}}) - \frac{1}{N_{\text{samp}}} \sum_{i=1}^{N_{\text{samp}}} \psi(N_{y_i}) + \psi(k) - \frac{1}{N_{\text{samp}}} \sum_{i=1}^{N_{\text{samp}}} \psi(m_i), \tag{10}$$

where $\psi(\cdot)$ denotes the digamma function, N_{y_i} is the count of training samples sharing class label y_i, k represents the neighborhood size in the joint continuous-discrete space ($k = 3$), and m_i is the count of data points whose marginal continuous distance to sample i along feature f_j is strictly smaller than the distance to its k-th neighbor in the class-conditioned subspace. To establish a standardized

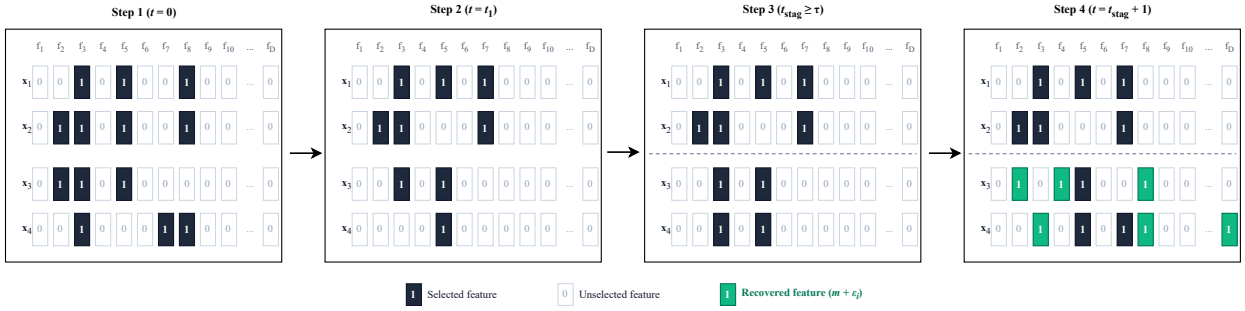


Figure 3: Process of prior-guided sparse initialization and adaptive recovery.

probability profile across heterogeneous datasets, these raw mutual information scores are normalized into a prior relevance vector $\mathbf{m} = [m_1, \ldots, m_D]^T \in [0,1]^D$:

$$m_j = \frac{\hat{I}(f_j; Y) - \min_{1 \leq k \leq D} \hat{I}(f_k; Y)}{\max_{1 \leq k \leq D} \hat{I}(f_k; Y) - \min_{1 \leq k \leq D} \hat{I}(f_k; Y) + \epsilon}, \quad j = 1, \ldots, D, \tag{11}$$

where $\epsilon > 0$ is a small numerical constant preventing division by zero.

To guarantee strict methodological integrity and eliminate selection bias, the filter prior calculation is architecturally restricted to the training partition presented to the optimizer. In an inductive machine learning deployment where an external unseen test partition $\mathcal{D}_{\text{test}}$ or an outer nested evaluation loop is prescribed, the prior relevance vector $\mathbf{m}$ and the subsequent evolutionary wrapper search are executed strictly on the training partition $\mathcal{D}_{\text{train}}$, ensuring that unseen validation samples remain completely unobserved throughout all search stages. In the standardized benchmark optimization protocol adopted in this work (Section 4.1), $\mathcal{D}$ serves as the complete benchmark problem instance, within which an internal 5-fold cross-validation scheme functions as the wrapper fitness evaluation surrogate.

To prevent population homogenization while guiding search trajectories toward informative dimensions, candidate particles are initialized through a hybrid seeding scheme parameterized by a seeding ratio $\rho \in (0,1)$. Continuous position vectors $\mathbf{p}_i = [p_{i,1}, \ldots, p_{i,D}] \in [0,1]^D$ are constructed as:

$$\mathbf{p}_i = \begin{cases} \text{clip}\left(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1\right), & 1 \leq i \leq \lfloor \rho N \rfloor, \\ \mathbf{u}_i, & \lfloor \rho N \rfloor < i \leq N, \end{cases} \tag{12}$$

where $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ represents zero-mean Gaussian perturbation with standard deviation σ, $\mathbf{u}_i \sim \mathcal{U}(0,1)^D$ provides uniform stochastic coverage across unguided dimensions, and $\text{clip}(z, 0, 1) = \max(0, \min(1, z))$ restricts continuous coordinates to the unit hypercube.

Continuous position vectors $\mathbf{p}_i$ are subsequently transformed into binary decision vectors $\mathbf{x}_i = [x_{i,1}, \ldots, x_{i,D}] \in \{0,1\}^D$ via a deterministic Heaviside step mapping with threshold $\theta \in (0.5, 1)$:

$$x_{i,j} = \begin{cases} 1, & \text{if } p_{i,j} \geq \theta, \\ 0, & \text{otherwise}, \end{cases} \quad j = 1, \ldots, D. \tag{13}$$

By setting the cut-off threshold $\theta > 0.50$, only features possessing above-average prior relevance are initialized in active states. This produces an intrinsically sparse initial population that biases exploration toward high-potential biomarker candidates, thereby resolving the cold-start bottleneck, reducing early evaluation latency, and accelerating convergence toward promising feature subspaces.

Estimation robustness and mitigation of the Winner's curse under sample scarcity: In ultra-high-dimensional and sample-scarce regimes where $D \gg 10^3$ and $N_{\text{samp}} \ll 100$ (such as *Adenoma* with $N_{\text{samp}} = 36$, $D = 7{,}399$ or *ALL4* with $N_{\text{samp}} = 63$, $D = 12{,}600$), non-parametric entropy estimation on limited observations inherently exhibits non-trivial finite-sample variance. Under extensive dimensional testing, stochastic sample alignment across thousands of candidate features may cause irrelevant noise dimensions to obtain spuriously elevated mutual information estimates—a classic manifestation of the Winner's curse in high-dimensional screening. If subjected to standard Min-Max normalization, such estimation fluctuations risk being amplified toward $m_j \approx 1$, potentially injecting false-positive biases into the initial population.

Crucially, KDS-EL establishes a multi-layered architectural defense that insulates the evolutionary search against noisy prior contamination. First, KDS-EL fundamentally differs from conventional filter pre-selectors by rejecting irreversible hard feature truncation; the normalized vector $\mathbf{m}$ acts solely as a continuous soft prior, leaving the entire $\mathbb{R}^D$ feature space fully accessible throughout search. Second, the hybrid initialization in (12) injects Gaussian dispersion $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ around $\mathbf{m}$ while maintaining an unguided sub-swarm $\mathbf{u}_i \sim \mathcal{U}(0,1)^D$ across $(1-\rho)N$ particles, preventing premature population collapse into false-positive empirical peaks. Third, every candidate subset is immediately subjected to wrapper cross-validation classification feedback; because spurious noise features fail to generalize across cross-validation folds, they generate zero or negative fitness differentials. Under the exponential memory decay governed by decay factor γ in (17), their transient weights in the cumulative importance matrix $\mathbf{W}$ rapidly decay to zero. Consequently, the knowledge-guided elite refinement operator swiftly purges these non-contributing dimensions in favor of synergistic, predictive biomarkers. Finally, should the swarm become temporarily trapped in empirical local attractors, the adaptive stagnation recovery routine (Section 3.2.2) re-samples the stagnant subpopulation under fresh stochastic perturbations, ensuring robust long-term exploratory self-correction.

3.2.2. *Adaptive stagnation recovery*

In multi-modal high-dimensional search spaces, population diversity inevitably decays across generations, which may lead the swarm into suboptimal attractors. To resolve this issue, KDS-EL incorporates an adaptive re-initialization routine triggered whenever the global-best fitness fails to improve for τ consecutive generations.

As depicted in Fig. 3, particles are first sorted in ascending order of their personal-best fitness values. The superior half of the swarm is retained intact to safeguard historical search trajectories and verified biomarker combinations. Concurrently, the lower-performing half is refreshed by re-sampling continuous positions around the prior relevance vector $\mathbf{m}$ under Gaussian perturbation:

$$\mathbf{p}_i \leftarrow \text{clip}\left(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1\right), \quad x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \geq \theta), \quad \forall j = 1, \ldots, D, \tag{14}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. The refreshed particles are immediately re-evaluated under current objective weights, and the stagnation counter is reset. By combining elite preservation with targeted prior re-injection, this mechanism restores exploratory diversity within high-potential subspaces without discarding accumulated optimization experience.

3.3. *Cumulative importance learning and elite refinement*

To overcome the search amnesia and destructive perturbation bottlenecks of conventional meta-heuristics, KDS-EL establishes a closed-loop knowledge learning mechanism comprising online importance accumulation and targeted directional elite refinement.

In ultra-high-dimensional feature selection ($D \gg 10^3$), traditional evolutionary wrappers rely on undirected stochastic bit flips with a uniform mutation probability $p_{\text{mut}} = 1/D$. Under this blind perturbation regime, meta-heuristics inevitably suffer from two fatal structural search lesions. First, when an established, highly informative biomarker ($x_{e,j} = 1$) is blindly mutated to 0, the delicate multi-feature synergy is abruptly dismantled, inducing severe classification degradation. Second, when an uninformative background noise feature ($x_{e,j} = 0$) is randomly flipped to 1, the candidate subset expands unnecessarily, escalating distance evaluation latency and precipitating classifier overfitting. Because the probability of hitting the few decisive dimensions within a 2^D combinatorial neighborhood is infinitesimal ($\mathcal{O}(1/D) \approx 10^{-4}$), undirected mutation squanders the vast majority of function evaluations on destructive trials. Furthermore, conventional meta-heuristics exhibit severe search amnesia, as historical fitness feedback from intermediate iterations is discarded immediately after particle position updates.

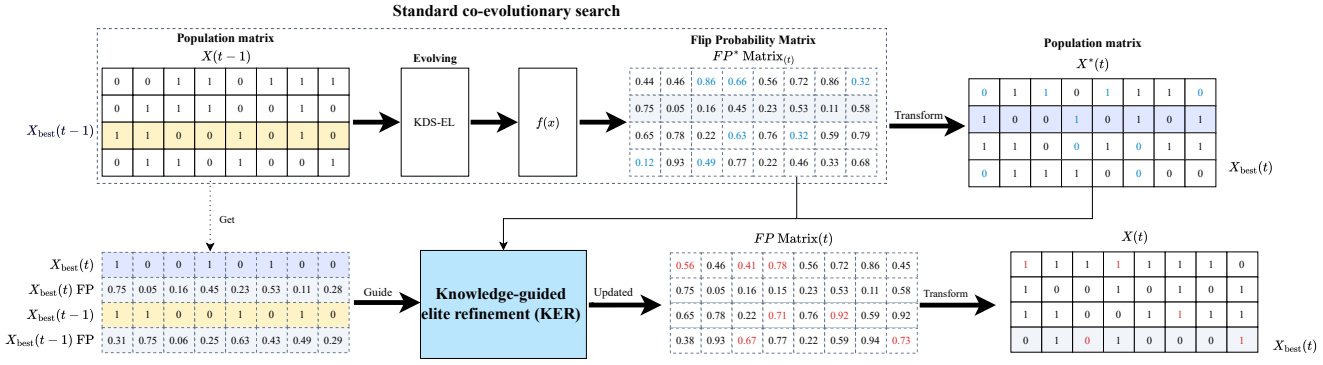


Figure 4: Process of the knowledge-guided elite refinement strategy.

3.3.1. Online feature importance learning

To systematically harvest search experience across generations, KDS-EL maintains a continuous cumulative feature-importance vector $\mathbf{W} = [W_1, \ldots, W_D]^T \in \mathbb{R}^D$, initialized to $\mathbf{0}$.

At generation t, whenever a candidate particle $\mathbf{x}_i^{(t)}$ achieves a strict fitness improvement over its historical personal best $\mathbf{p}_{\text{best},i}$, the differential feature set $C_i^{(t)}$ comprising features whose selection status changed is extracted:

$$C_i^{(t)} = \left\{ j \in \{1, \ldots, D\} \;\middle|\; x_{i,j}^{(t)} \neq p_{\text{best},i,j} \right\}. \tag{15}$$

Let $\delta_i^{(t)} = f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) - f(\mathbf{x}_i^{(t)}, \mathbf{w}(t)) > 0$ denote the scalar fitness improvement. Each altered feature $j \in C_i^{(t)}$ receives a directional reward proportional to the magnitude of $\delta_i^{(t)}$:

$$W_j \leftarrow W_j + \delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1), \quad \forall j \in C_i^{(t)}, \tag{16}$$

where the term $(2x_{i,j}^{(t)} - 1)$ yields $+1$ if feature j was newly activated and -1 if deactivated. Candidate solutions that fail to improve fitness ($\delta_i^{(t)} \leq 0$) leave $\mathbf{W}$ unchanged.

To prevent early-discovered features from dominating the importance landscape indefinitely and maintain sensitivity toward emergent synergistic biomarker interactions, an exponential memory decay is applied at the conclusion of each generation:

$$\mathbf{W} \leftarrow \gamma \cdot \mathbf{W}, \tag{17}$$

where $\gamma \in (0, 1)$ is the memory retention factor. Consequently, features that consistently contribute to fitness improvements sustain high positive scores, whereas transiently selected or uninformative dimensions exponentially fade toward zero.

3.3.2. Knowledge-guided elite refinement

Elite individuals represent the most promising feature subsets discovered by the swarm; however, they may still retain redundant dimensions or lack subtle synergistic biomarkers. Rather than blind perturbation, the KER operator exploits the continuous importance landscape $\mathbf{W}$ to guide local mutation toward high-potential dimensions under bounded computation ($k_e \ll N$).

At each generation, the top k_e individuals ranked by fitness are selected into the elite pool. For each elite candidate $\mathbf{x}_e \in \{0, 1\}^D$, the number of perturbed features is set to $k_{\text{flip}} = \max(1, \lfloor \phi D \rfloor)$, where $\phi \in (0, 1)$ represents the perturbation fraction. As depicted in the data-flow diagram in Fig. 4, KER probabilistically executes one of two operational pathways:

Branch 1: Ergodic random exploration (p_{rand}): To preserve search ergodicity and prevent the elite pool from stagnating in deceptive local optima, a feature index subset $\mathcal{S}_{\text{rand}} \subset \{1, \ldots, D\}$ of size k_{flip} is sampled uniformly at random, and the corresponding bits are inverted:

$$x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{rand}}, \quad |\mathcal{S}_{\text{rand}}| = k_{\text{flip}}. \tag{18}$$

Branch 2: Knowledge-guided directional refinement ($1 - p_{\text{rand}}$): For each feature $j \in \{1, \ldots, D\}$, a state-dependent directional flip score σ_j is computed:

$$\sigma_j = (1 - 2x_{e,j}) \cdot W_j = \begin{cases} +W_j, & \text{if } x_{e,j} = 0, \\ -W_j, & \text{if } x_{e,j} = 1. \end{cases} \tag{19}$$

Algorithm 1 Knowledge-guided elite refinement

Require: Elite candidate solution $\mathbf{x}_e \in \{0, 1\}^D$, cumulative importance vector $\mathbf{W} \in \mathbb{R}^D$, perturbation fraction $\phi \in (0, 1)$, exploration probability $p_{\text{rand}} \in [0, 1]$, dynamic weight vector $\mathbf{w}(t) \in [0, 1]^2$
Ensure: Refined elite candidate solution $\mathbf{x}_e^\star \in \{0, 1\}^D$

```
k_flip ← max(1, ⌊φ · D⌋)
x'_e ← x_e                                   ▷ Initialize mutant candidate
Sample random variable r ~ U(0, 1)
if r < p_rand then                           ▷ Branch 1: Ergodic random exploration
    Sample random feature index subset S_rand ⊂ {1, ..., D} with |S_rand| = k_flip
    x'_{e,j} ← 1 − x_{e,j},   ∀j ∈ S_rand
else                                         ▷ Branch 2: Knowledge-guided directional refinement
    for j = 1 to D do
        σ_j ← (1 − 2x_{e,j}) · W_j           ▷ Compute directional flip score via (19)
    end for
    Identify index subset S_know of size k_flip with top σ_j scores via (20)
    x'_{e,j} ← 1 − x_{e,j},   ∀j ∈ S_know
end if
Evaluate candidate fitness f(x'_e, w(t)) via (23)
if f(x'_e, w(t)) < f(x_e, w(t)) then         ▷ Greedy 1-to-1 in-place replacement
    x*_e ← x'_e
else
    x*_e ← x_e
end if
return x*_e
```

A positive flip score $\sigma_j > 0$ indicates either an unselected feature ($x_{e,j} = 0$) with positive historical contribution ($W_j > 0$), or an active feature ($x_{e,j} = 1$) whose deactivation has historically improved fitness ($W_j < 0$). The feature subset $\mathcal{S}_{\text{know}}$ containing the k_{flip} features with the largest directional flip scores is identified:

$$\mathcal{S}_{\text{know}} = \arg \max_{\substack{\mathcal{S} \subseteq \{1, \ldots, D\} \\ |\mathcal{S}| = k_{\text{flip}}}} \sum_{j \in \mathcal{S}} \sigma_j, \tag{20}$$

and their binary states are inverted: $x'_{e,j} \leftarrow 1 - x_{e,j}, \forall j \in \mathcal{S}_{\text{know}}$.

Greedy one-to-one in-place replacement: The perturbed candidate $\mathbf{x}'_e$ is evaluated using the objective function $f(\mathbf{x}'_e, \mathbf{w}(t))$. A greedy acceptance rule determines solution admission:

$$\mathbf{x}_e^\star = \begin{cases} \mathbf{x}'_e, & \text{if } f(\mathbf{x}'_e, \mathbf{w}(t)) < f(\mathbf{x}_e, \mathbf{w}(t)), \\ \mathbf{x}_e, & \text{otherwise.} \end{cases} \tag{21}$$

Upon achieving a strict fitness improvement, $\mathbf{x}'_e$ immediately replaces $\mathbf{x}_e$ in the population and updates the global-best solution if $f(\mathbf{x}'_e, \mathbf{w}(t)) < f_{\text{gbest}}$. Because the elite pool size k_e is fixed ($k_e \ll N$), KER operates within an exact bounded overhead of k_e evaluations per generation. The complete procedural logic of KER is formalized in Algorithm 1.

Dual-scale search dynamics and dimension-scaling analysis: The perturbation budget $k_{\text{flip}} = \max(1, \lfloor \phi D \rfloor)$ scales linearly with ambient dimensionality D, producing distinct search behaviors across problem scales. In moderate-dimensional domains where $D \leq 2{,}000$, k_{flip} produces small-scale bit flips that execute fine-grained local neighborhood refinement around elite solutions. Conversely, in ultra-high-dimensional spaces where $D > 10^4$—such as the 54,675-dimensional *T1D* benchmark where $k_{\text{flip}} \approx 546$—KER shifts dynamically from localized perturbation into a guided macro-recombination operator.

Operating at this broader structural scale allows the swarm to evaluate coordinated multi-feature activations and deactivations guided by $\mathbf{W}$, facilitating transitions across distant combinatorial basins. Crucially, the greedy one-to-one in-place replacement rule in (21) serves as an intrinsic safeguard against destructive mutations: whenever an extensive macro-flip degrades classification accuracy or inflates feature penalties, the candidate mutant is discarded without modifying the elite pool. While fixed-fraction scaling $\lfloor \phi D \rfloor$ performs consistently across diverse benchmark scales, adaptively modulating the perturbation budget according to the active subset cardinality $d_{\text{selected}} = \|\mathbf{x}_e\|_0$ via $k_{\text{flip}} = \max(1, \lfloor \phi \cdot \|\mathbf{x}_e\|_0 \rfloor)$, or enforcing an upper saturation cap $k_{\max}$, represents an effective adaptive refinement for extreme-dimensional applications.

3.4. Dynamic two-phase weight scheduling

In high-dimensional feature selection, wrapper optimization must balance two fundamentally conflicting objectives: minimizing classification error and minimizing the selected feature subset cardinality. In the early evolutionary phase, the search space is dominated by an overwhelming number of redundant and noisy dimensions; thus, the primary optimization imperative is aggressive dimensionality pruning. In contrast, during the late evolutionary phase when search trajectories have converged to a compact subspace, the priority shifts to fine-tuning synergistic biomarker interactions to maximize classification accuracy. Conventional static weighting schemes with fixed trade-off coefficients fail to reconcile these temporally distinct demands, frequently inducing either inadequate early feature elimination or premature penalization of critical predictive features in later stages.

To resolve this temporal conflict, KDS-EL introduces the dynamic TPS strategy, which decouples the evolutionary horizon into two distinct functional phases. The total generation budget T is divided at a split point parameterized by the phase transition ratio $\kappa \in (0, 1)$. The dynamic objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)] \in [0, 1]^2$, satisfying $w_a(t) + w_f(t) = 1$, is formulated as:

$$\mathbf{w}(t) = \begin{cases} [w_{a,1}, w_{f,1}], & 1 \le t \le \lfloor \kappa T \rfloor, \\ [w_{a,2}, w_{f,2}], & \lfloor \kappa T \rfloor < t \le T, \end{cases} \tag{22}$$

where the weight coefficients satisfy the ordering $w_{f,1} > w_{f,2}$ and $w_{a,1} < w_{a,2}$.

At generation t, the unified scalar fitness $f(\mathbf{x}, \mathbf{w}(t))$ of a binary feature subset $\mathbf{x} \in \{0, 1\}^D$ is evaluated as:

$$f(\mathbf{x}, \mathbf{w}(t)) = w_a(t) \cdot (1 - \text{Acc}(\mathbf{x})) + w_f(t) \cdot \left(\frac{\|\mathbf{x}\|_0}{D} \right), \tag{23}$$

where $\text{Acc}(\mathbf{x}) \in [0, 1]$ represents the 5-fold cross-validation classification accuracy evaluated by a k-NN classifier, and $\|\mathbf{x}\|_0 = \sum_{j=1}^{D} x_j$ denotes the ℓ_0-norm quantifying the number of active features.

During Phase 1 ($1 \le t \le \lfloor \kappa T \rfloor$), the elevated feature penalty $w_{f,1} = 0.30$ enforces severe selective pressure against non-informative features, rapidly compressing the active candidate dimensionality to a compact fraction of D. Subsequently, upon entering Phase 2 ($\lfloor \kappa T \rfloor < t \le T$), the weight vector transitions to prioritize classification accuracy ($w_{a,2} = 0.95$), empowering the swarm to explore subtle multi-gene synergies over the condensed feature subspace without being prematurely penalized for retaining essential predictive biomarkers.

3.5. Overall algorithmic procedure and complexity analysis

Algorithm 2 synthesizes the overall execution pipeline of KDS-EL into a structured multi-stage optimization framework. The complete workflow coordinates offline filter relevance extraction, knowledge-informed hybrid seeding, dynamic two-phase objective scheduling, collaborative dual-subpopulation search, differential importance harvesting, and targeted elite refinement.

3.5.1. Computational complexity analysis

The computational overhead of KDS-EL decouples into two distinct phases: one-time offline preprocessing and online generational evolution. In the offline phase, evaluating non-parametric mutual information scores across D candidate dimensions with N_{samp} training samples via k-nearest neighbors search requires $O(D \cdot N_{\text{samp}} \log N_{\text{samp}})$ operations, while normalizing and sorting these relevance estimates incurs $O(D \log D)$ time. Because this procedure is executed exactly once prior to the evolutionary search, the offline preprocessing complexity $\mathcal{T}_{\text{pre}} = O(D \cdot N_{\text{samp}} \log N_{\text{samp}} + D \log D)$ represents a negligible fraction of the total computational budget, especially in small-sample regimes where $N_{\text{samp}} \ll 100$. Throughout online evolution, each generation $t \in \{1, \ldots, T\}$ orchestrates three coordinated operational stages. First, subpopulation updates and cross-task knowledge transfer—comprising continuous velocity-position updates in $\mathcal{P}_1$ and discrete probabilistic sampling in $\mathcal{P}_2$—require $O(N \cdot D)$ arithmetic operations per generation. Second, wrapper

Algorithm 2 The overall framework of KDS-EL

Require: Training dataset $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$, population size $N \in \mathbb{Z}_+$, maximum generations $T \in \mathbb{Z}_+$, binarization threshold $\theta \in (0, 1)$, prior seeding ratio $\rho \in (0, 1)$, perturbation scale $\sigma \in \mathbb{R}_+$, forgetting factor $\gamma \in (0, 1)$, elite pool size $k_e \in \mathbb{Z}_+$, perturbation fraction $\phi \in (0, 1)$, random branch probability $p_{\text{rand}} \in [0, 1]$, phase transition split ratio $\kappa \in (0, 1)$, stagnation threshold $\tau \in \mathbb{Z}_+$

Ensure: Optimal compact feature subset $\mathbf{x}^\star \in \{0, 1\}^D$ and optimal fitness $f_{\text{gbest}} \in \mathbb{R}$

Stage 1: Filter-level relevance extraction

Compute normalized mutual information vector $\mathbf{m} \in [0, 1]^D$ via (10) and (11)

Stage 2: Knowledge-informed hybrid population seeding

Initialize population $\mathcal{P} = \{\mathbf{x}_i\}_{i=1}^{N}$ with $\lfloor \rho N \rfloor$ Gaussian MI seeds and $(N - \lfloor \rho N \rfloor)$ uniform particles via (12)–(13)

Initialize importance vector $\mathbf{W} \leftarrow \mathbf{0} \in \mathbb{R}^D$, stagnation counter $c_{\text{stall}} \leftarrow 0$, and initial weights $\mathbf{w}(1) = [w_{a,1}, w_{f,1}]$ via (22)

Evaluate initial candidate fitness values $\{f(\mathbf{x}_i, \mathbf{w}(1))\}_{i=1}^{N}$ via (23); initialize personal bests $\{\mathbf{p}_{\text{best},i}\}_{i=1}^{N}$ and global best $\mathbf{g}_{\text{best}}$

Stage 3: Dynamic two-phase co-evolutionary loop

for $t = 1$ **to** T **do**

  Update dynamic objective weight vector $\mathbf{w}(t)$ according to (22)

  Update continuous Subpopulation $\mathcal{P}_1$ via continuous PSO with inter-task knowledge transfer [12]

  Update discrete Subpopulation $\mathcal{P}_2$ via discrete probability sampling guided by $\mathbf{W}$

  Evaluate candidate solution fitness $\{f(\mathbf{x}_i, \mathbf{w}(t))\}_{i=1}^{N}$ using 5-fold cross-validation k-NN classifier via (23)

  Update cumulative importance vector $\mathbf{W}$ with improving differentials and apply memory decay γ via (16)–(17)

  Update personal bests $\{\mathbf{p}_{\text{best},i}\}_{i=1}^{N}$ and identify top k_e elite candidate indices $\mathcal{E}$

  for each $e \in \mathcal{E}$ **do** ▷ Knowledge-guided elite refinement

    $\mathbf{p}_{\text{best},e} \leftarrow \text{KER}(\mathbf{p}_{\text{best},e}, \mathbf{W}, \phi, p_{\text{rand}}, \mathbf{w}(t))$ ▷ Invoke Algorithm 1

  end for

  if $\min_i f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) < f(\mathbf{g}_{\text{best}}, \mathbf{w}(t))$ **then**

    Update global best $\mathbf{g}_{\text{best}} \leftarrow \arg\min_i f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t))$ and reset $c_{\text{stall}} \leftarrow 0$

  else

    Increment stagnation counter $c_{\text{stall}} \leftarrow c_{\text{stall}} + 1$

  end if

  if $c_{\text{stall}} \ge \tau$ **then** ▷ Adaptive stagnation recovery

    Refresh lower-performing half of population using Gaussian-perturbed MI seeds via (14)

    Re-evaluate refreshed particles and reset $c_{\text{stall}} \leftarrow 0$

  end if

end for

return $\mathbf{x}^\star \leftarrow \mathbf{g}_{\text{best}}$, $f_{\text{gbest}} \leftarrow f(\mathbf{g}_{\text{best}}, \mathbf{w}(T))$

fitness evaluation constitutes the dominant computational expenditure: for each candidate solution $\mathbf{x}_i$ with active feature cardinality $d_i = \|\mathbf{x}_i\|_0$, computing pairwise Euclidean distances under 5-fold cross-validation with a k-NN classifier incurs $O(N_{\text{samp}}^2 \cdot d_i)$ runtime, aggregating to $O(N \cdot N_{\text{samp}}^2 \cdot \bar{d}^{(t)})$ across the swarm of size N, where $\bar{d}^{(t)} = \frac{1}{N} \sum_{i=1}^{N} d_i$ denotes the average active feature count at generation t. Third, cumulative importance learning and elite refinement introduce targeted search intensification within strictly bounded overhead: extracting differential feature masks and updating the continuous memory vector $\mathbf{W}$ consumes $O(N \cdot D)$ operations; computing directional flip scores across all D dimensions and isolating the top k_{flip} features incurs $O(D \log D)$ time via partial quick-sort; and re-evaluating the refined elite candidates requires $O(k_e \cdot N_{\text{samp}}^2 \cdot d_{\text{elite}})$ operations. Because the elite pool size is strictly bounded with $k_e \ll N$ ($k_e = 3$), this refinement cost represents a marginal constant fraction of the overall generational budget. Accumulating these operational costs across T generations yields the total asymptotic time complexity of KDS-EL:

$$\mathcal{T}_{\text{total}} = O\left(T \cdot N \cdot N_{\text{samp}}^2 \cdot \bar{d}_{\text{selected}} + T \cdot N \cdot D \right), \tag{24}$$

where $\bar{d}_{\text{selected}}$ denotes the average active feature count across all evolutionary generations.

While KDS-EL exhibits the same asymptotic Big-O order as standard PSO-based feature selectors ($O(TNN_{\text{samp}}^2 D)$), it achieves an empirical wall-clock speedup exceeding 3.4× in practice. This dramatic efficiency gain originates directly from the prior-guided sparse initialization and dynamic TPS strategy: rather than evaluating dense subsets ($\bar{d} \approx D/2 \approx 5{,}000+$), KDS-EL aggressively compresses candidate dimensionality to $\bar{d}_{\text{KDS-EL}} \approx 80 \sim 150$ active features starting from the earliest generations. Because the k-NN distance computation cost scales strictly linearly with the number of active features d, compressing the active dimension by two orders of magnitude reduces the actual floating-point operation volume per evaluation by over 95%, translating theoretical sparsity into profound real-world computational savings.

From the perspective of memory consumption, the spatial footprint of

KDS-EL comprises the binary population matrix $\mathbf{X} \in \{0,1\}^{N\times D}$, historical personal best positions $\mathbf{P}_{\text{best}} \in \{0,1\}^{N\times D}$, the continuous feature importance memory vector $\mathbf{W} \in \mathbb{R}^{D}$, and the benchmark training dataset $\mathcal{D} \in \mathbb{R}^{N_{\text{samp}}\times D}$. The total spatial complexity is $\mathcal{S}_{\text{total}} = \mathcal{O}\big((2N + N_{\text{samp}} + 1)D\big) = \mathcal{O}\big((N + N_{\text{samp}})D\big)$, which scales strictly linearly with feature dimensionality D. Because memory allocation is dominated by standard contiguous floating-point and integer arrays, the required working memory remains on the order of megabytes even for ultra-high-dimensional datasets with $D > 10^5$, ensuring seamless execution on commodity computational hardware without out-of-memory overhead.

4. Experimental results and discussion

This section outlines the experimental framework and presents a comprehensive empirical evaluation of KDS-EL across eight analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects. All aggregate numbers are computed over the 20 benchmark datasets described in Section 4.1.

4.1. Experimental setup

This subsection outlines the experimental framework designed to evaluate the classification performance, dimensionality reduction capability, computational efficiency, and hyperparameter robustness of KDS-EL.

4.1.1. High-dimensional benchmark datasets

To validate the performance and generalization ability of KDS-EL across diverse problem domains, a total of 20 benchmark datasets widely adopted in the evolutionary feature selection literature [8, 11, 12] are selected. These datasets encompass diverse categories and sample sizes across biomedical gene-expression microarrays, text categorization corpora, image recognition, speech processing, and synthetic benchmarks. Such diverse data distributions and dimensional complexities effectively evaluate the generalization capability and feature reduction efficacy of feature selection algorithms under challenging conditions. The detailed characteristics of the 20 benchmark datasets are summarized in Table 2.

4.1.2. Comparison algorithms

We select two primary categories of competitor algorithms for empirical benchmark analysis alongside the proposed KDS-EL framework, as systematically categorized in Table 1. The first category comprises established single-objective evolutionary algorithms in high-dimensional feature selection: FW-PSO [8], PSO-CSM [11], SaWDE [9], IGPSO [10], and BPSO [7]. The second category encompasses the foundational multi-task precursor MEL [12] alongside state-of-the-art single-objective meta-heuristics published between 2023 and 2024: the evolutionary filter approach DENCA-SO [14], the binary frost ice optimizer bIRIME [13], and the binary crayfish optimization algorithm BinCOA [15]. The corresponding algorithmic hyperparameter settings are summarized in Table 3. In addition, two constituent ablation variants of KDS-EL (KDS-EL w/o TPS and KDS-EL w/o TPS&KER) are investigated in Section 4.2.5 to isolate and quantify individual component contributions. In total, this 12-method comparison suite systematically spans canonical swarm heuristics, evolutionary filter paradigms, and contemporary state-of-the-art designs, guaranteeing an exhaustive, rigorous, and reproducible empirical evaluation.

4.1.3. Evaluation metrics and statistical test protocols

Algorithm performance is evaluated across three primary quantitative criteria: mean classification accuracy alongside standard deviation, the number of selected features alongside the feature selection ratio (FSR = $\frac{\bar{d}}{D} \times 100\%$), and computational runtime in seconds. The k-nearest neighbor classifier with $k = 3$ serves as the base evaluation model due to its non-parametric nature, parameter-free reproducibility, and high sensitivity to candidate feature subset discriminability. Following the standard benchmarking protocol established in high-dimensional evolutionary feature selection literature [8, 11, 12], candidate feature subsets are evaluated using 5-fold stratified cross-validation as the internal wrapper fitness surrogate across all problem instances. Under this standardized formulation, each benchmark dataset $\mathcal{D}$ is treated as a self-contained combinatorial optimization instance; the 5-fold cross-validation classification accuracy guides the evolutionary search trajectory and evaluates the discriminative quality of the optimal feature subsets under identical computational budgets across all competing methods.

To guarantee experimental fairness and statistical reliability, every algorithm is executed with a population size of $N = 20$ candidate solutions over $T_{\max} = 100$ generations across 10 independent runs with distinct random seeds. In ultra-high-dimensional wrapper feature selection, each fitness evaluation requires training and validating a classifier across thousands or tens of thousands of candidate dimensions (e.g., up to 54,675 dimensions in *T1D*), imposing an exceptionally intensive computational burden of $N \times T_{\max} = 2{,}000$ cross-validation evaluations per single run. Executing 10 independent runs per dataset corresponds to 50 distinct stratified cross-validation folds, which provides an adequate sample size to capture classifier generalization variance and algorithmic stability while remaining computationally tractable across an extensive 12-method comparison suite. This 10-run protocol strictly aligns with the benchmark evaluation standard established in recent top-tier literature on ultra-high-dimensional feature selection, most notably the multi-task evolutionary learning framework MEL published in *IEEE Transactions on Knowledge and Data Engineering* [12], thereby guaranteeing an equitable and reproducible empirical comparison.

Furthermore, to determine whether the empirical advantages of KDS-EL over competing paradigms are statistically significant rather than stochastic artifacts, we strictly follow the multi-problem non-parametric statistical testing framework recommended by Demšar [39]. In this protocol, statistical power is governed by the number of independent benchmark problems ($N_{\text{datasets}} = 20$), with each problem represented by the paired summary performance across the independent runs. Specifically, the non-parametric Friedman test [40] is utilized to establish global omnibus algorithm rankings across all 20 benchmark datasets. Pairwise Wilcoxon signed-rank tests at the $\alpha = 0.05$ significance level assess directional superiority between KDS-EL and each competitor, with the Holm–Bonferroni step-down procedure [41] applied to rigorously control the family-wise error rate across multiple hypothesis comparisons.

Table 2: Characteristics of the 20 benchmark datasets.

No.	Dataset	Samples (n)	Features (D)	Classes (C)	Domain
1	Adenoma	36	7,457	2	Gene expression
2	ALL_AML	72	7,129	2	Gene expression
3	ALL1	128	12,625	2	Gene expression
4	ALL4	93	12,625	2	Gene expression
5	Colon	62	2,000	2	Gene expression
6	DLBCL	77	7,129	2	Gene expression
7	Gastric	65	22,645	2	Gene expression
8	Gastric1	144	22,283	2	Gene expression
9	Gastric2	124	22,283	2	Gene expression
10	Leukaemia	72	7,129	2	Gene expression
11	Lymphoma	45	4,026	2	Gene expression
12	Myeloma	173	12,625	2	Gene expression
13	Prostate	102	12,625	2	Gene expression
14	SRBCT	83	2,308	4	Gene expression
15	T1D	101	54,675	2	Gene expression
16	BASEHOCK	1,993	4,862	2	Text
17	PCMAC	1,943	3,289	2	Text
18	COIL20	1,440	1,024	20	Image
19	ISOLET	1,560	617	26	Speech
20	madelon	2,600	500	2	Synthetic

4.2. Experimental analysis

This subsection presents a comprehensive empirical evaluation of KDS-EL across eight complementary analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects.

4.2.1. Comparison of classification accuracy

The KDS-EL framework was systematically evaluated alongside 11 competitor algorithms and ablative variants across 20 high-dimensional benchmark datasets. Table 4 presents the experimental results for average classification accuracy and standard deviation, with the best values in boldface and the second-best underlined.

First, comparing KDS-EL to all 11 comparison methods overall, KDS-EL achieves the highest average classification accuracy of 0.9574 across the 20 benchmark datasets, securing the top performance across the suite. Specifically, on ultra-high-dimensional microarrays characterized by severe sample sparsity,

Table 3: Relevant parameter settings for all algorithms.

Algorithm / Component	Parameter setting
Common experimental parameters:	
Population size N	20
Maximum generations $T_{\max}$	100
Classifier	3-NN with 5-fold stratified CV
Binarization threshold θ	0.60
Independent runs	10
Proposed KDS-EL framework:	
Subpopulations	$N_1 = 10,\ N_2 = 10$
Inertia weight ω	0.90
Acceleration coefficients c_1, c_2, c_3	2.0, 2.0, 2.0
Knowledge decay factor γ	0.99
Perturbation fraction ϕ	0.01
Elite pool size k_e	3
Random exploration rate p_{rand}	0.10
MI initialization ratio ρ	0.50
Stagnation threshold τ	20 generations
Phase split ratio κ	0.70
Phase-1 weights $\mathbf{w}_1$	[0.70, 0.30]
Phase-2 weights $\mathbf{w}_2$	[0.95, 0.05]
Ablative variants:	
KDS-EL w/o TPS	Static weights $\mathbf{w}_s = [0.90, 0.10]$
KDS-EL w/o TPS&KER	Without KER and TPS, $\mathbf{w}_s = [0.90, 0.10]$
Original MEL [12]	Uniform init, $\mathbf{w}_s = [0.90, 0.10]$
Competitor algorithms:	
FWPSO [8]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ \alpha_{\text{sa}} = 0.95$
PSO-CSM [11]	$\omega = 0.7298,\ c_1 = c_2 = 1.49618,\ p_m = 0.05$
SaWDE [9]	$F \in [0.1, 0.9],\ CR \in [0.1, 0.9]$
IGPSO [10]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0$, IG scoring
BPSO [7]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ V_{\max} = 6.0$
DENCA-SO [14]	$F = 0.5,\ CR = 0.2,\ W = 0.05$, NCFS $\lambda = 0.01$
bIRIME [13]	S-shaped transfer, CMS mutation, restart limit $L = 5$
BinCOA [15]	ROBL initialization, Sigmoid transfer, crisscross (HC/VC)

KDS-EL establishes decisive performance advantages, reaching classification accuracies of 0.9522 on the 12,625-dimensional *ALL4*, 0.9500 on *Colon*, and 0.8598 on the 54,675-dimensional *T1D*, consistently demonstrating superior discriminative capacity over competing algorithms. Secondly, compared to established single-objective evolutionary algorithms such as FWPSO, PSO-CSM, SaWDE, IGPSO, and BPSO, KDS-EL elevates overall mean accuracy from the 0.8698–0.9177 range to 0.9574, confirming that closed-loop knowledge transfer and Two-Phase Weight Scheduling effectively overcome the random exploratory drift of standard swarm operators. Crucially, against the three recent top-tier single-objective competitors (2023–2024), KDS-EL consistently demonstrates superior predictive capabilities. Specifically, KDS-EL surpasses the recent binary frost ice optimizer bIRIME [13] (0.9454) by a statistically significant margin of 1.20 percentage points ($p = 0.0296$), exceeds the binary crayfish optimizer BinCOA [15] (0.8973) by 6.01 percentage points ($p = 9.54 \times 10^{-6}$), and outpaces the evolutionary filter approach DENCA-SO [14] (0.8712) by 8.62 percentage points ($p = 9.54 \times 10^{-6}$). Furthermore, compared with the foundational precursor MEL algorithm (0.8868), KDS-EL achieves an accuracy improvement of 7.06 percentage points ($p = 1.91 \times 10^{-6}$), demonstrating that mutual information initialization and knowledge-guided elite refinement fundamentally enhance the algorithm's biomarker discovery capability and localized exploitation. Overall, KDS-EL discovers superior feature subsets that yield substantially higher classification accuracy.

A closer examination of individual datasets and algorithmic dispersion provides further insights into the search behavior and robustness of KDS-EL. On the few datasets where KDS-EL does not record the nominal highest accuracy, such as *COIL20* and *ISOLET*, competitors such as bIRIME achieve marginal gains (e.g., 0.9112 vs. 0.8894 on *ISOLET*) only by retaining more than double the feature count (133 vs. 60 features), whereas KDS-EL attains competitive accuracy while preserving extreme feature sparsity. On saturated ceiling-effect datasets such as *ALL1*, *Adenoma*, and *SRBCT*, KDS-EL attains 0.9998, 0.9996, and 0.9986 accuracy with minimal signatures of 39, 66, and 64 features, respectively. Furthermore, KDS-EL demonstrates remarkably low run-to-run dispersion across independent executions, confirming that hybrid mutual information initialization and knowledge-guided elite refinement not only deliver higher accuracy in identifying optimal biomarker subsets, but also substantially enhance algorithmic stability and reproducibility.

4.2.2. Comparison of the number of selected features

The KDS-EL framework was evaluated against 11 comparison methods across 20 high-dimensional benchmark datasets. Table 5 presents the experimental results for the average number of selected features and the feature selection ratio, with the optimal values in bold.

First, comparing KDS-EL to the comparison methods overall, KDS-EL achieves the most compact feature subsets across the entire benchmark suite, selecting an overall mean of only 87.8 features and compressing the search space down to an average selection ratio of approximately 1.1%. This extreme dimensionality reduction capability is especially pronounced on challenging biomedical microarrays such as *ALL1*, *Colon*, *Gastric1*, and *Prostate*, where KDS-EL isolates minimal biomarker signatures of fewer than 40 genes—notably 14 on *Colon*, 31 on *Gastric1*, and 26 on *Prostate*—while delivering near-perfect classification accuracies between 0.9500 and 0.9998. Secondly, compared to specialized single-objective evolutionary algorithms, even the most compact competitor, FWPSO, retains an average of 1,068.9 features—over 12 times the subset size of KDS-EL—demonstrating that Two-Phase Weight Scheduling successfully enforces aggressive feature elimination during early iterations. Against the precursor MEL method with 2,722.2 features, KDS-EL achieves a remarkable 31-fold reduction in feature count, confirming that mutual information initialization and knowledge-guided elite refinement effectively steer particles into highly sparse, discriminative subspaces rather than dense noisy regions. Crucially, when contrasted with recent single-objective competitors, the feature-sparsity advantage of KDS-EL is overwhelming. Recent top-tier competitor bIRIME selects an average of 407.7 features—over 4.6 times more features than KDS-EL—because its multi-candidate trial mutations lack targeted importance decay guidance. Meanwhile, algorithms such as BinCOA (5,420.7 features, 48.3% selection ratio), DENCA-SO (5,323.4 features, 48.2% selection ratio), and standard BPSO (5,296.5 features, 47.0% selection ratio) remain completely trapped in the uninformative 50% dimensionality attractor, selecting over 5,200 redundant features. This occurs because unguided binarization without proactive pruning pressure stochastically centers active feature bits around $D/2$. In stark contrast, KDS-EL achieves an average feature selection ratio of merely 1.1%, compressing high-dimensional spaces by over 98% while securing superior classification performance.

From the perspective of algorithmic components, the step-wise performance progression reveals an aggregate, macro-level compression in feature redundancy across the benchmark suite. Averaged over all 20 benchmark datasets, the feature count contracts from 2,722.2 in the precursor MEL algorithm to 290.5 with mutual information seeding, further shrinks to 121.7 through knowledge-guided elite refinement, and ultimately settles at an overall compact signature of 87.8 features under the dynamic TPS strategy. This hierarchical reduction demonstrates that each proposed mechanism eliminates non-informative dimensions in a complementary manner, establishing clear overall feature-compression superiority over existing evolutionary wrappers.

Crucially, while feature reduction progresses consistently in terms of the multi-dataset average, subset cardinality on individual problem instances reflects a deliberate, domain-specific trade-off governed by the two-phase scheduling dynamics. On several challenging biomedical microarrays, such as *Adenoma*, *Lymphoma*, and *SRBCT*, the complete KDS-EL framework retains moderately more features than the static-weight variant without TPS. This behavior directly manifests the underlying design rationale of the two-phase strategy: upon transitioning to Phase 2, the feature cardinality penalty $w_{f,2} = 0.05$ is relaxed while the classification accuracy weight $w_{a,2} = 0.95$ is prioritized. Rather than enforcing an inflexible sparsity constraint that risks pruning decisive biomarker interactions, this controlled relaxation allows the search engine to strategically retrieve a small subset of weakly individual but strongly synergistic features that resolve ambiguous classification boundaries. Consequently, this targeted trade-off converts marginal feature retention into a statistically confirmed accuracy advantage across the benchmark suite ($p = 6.29 \times 10^{-5}$ under the Holm–Bonferroni procedure), confirming that KDS-EL balances extreme feature sparsity with optimal predictive discrimination.

4.2.3. Fitness value and convergence analysis

The scalar fitness value formalizes the joint optimization trade-off between minimizing classification error and compressing the active feature ratio throughout search, directly capturing each algorithm's exploratory efficiency and exploitation depth. To evaluate convergence dynamics comprehensively and avoid selective benchmark bias, Fig. 5 plots the mean fitness trajectories across all 20 benchmark datasets over 100 iterations for KDS-EL and four representative single-objective wrapper competitors: FWPSO, SaWDE, MEL, and classical BPSO.

As demonstrated across the 20 panels in Fig. 5, KDS-EL establishes a de-

Table 4: Comparison of pure classification accuracy (mean ± standard deviation across 10 independent runs) on 20 benchmark datasets. Best results are in **boldface**, and second-best are underlined. Wilcoxon signed-rank symbols +, ≈, and − denote significant superiority, parity, or inferiority of the proposed KDS-EL at $\alpha = 0.05$. The bottom rows summarize macro-average accuracy and directional win/tie/loss counts (+/ ≈ /−).

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
ALL1	0.9998	0.9996 $^+$	0.9996 $^+$	0.9664 $^+$	**1.0000** $^-$	0.9952 $^+$	0.9900 $^+$	**1.0000** $^-$	0.9710 $^+$	0.9802 $^+$	**1.0000** $^-$	0.9534 $^+$
ALL4	**0.9522**	0.9418 $^\approx$	0.9415 $^+$	0.8704 $^+$	0.9140 $^+$	0.8617 $^+$	0.8363 $^+$	0.7634 $^+$	0.8062 $^+$	0.7914 $^+$	0.7043 $^+$	0.7569 $^+$
ALL_AML	0.9995	0.9990 $^+$	0.9886 $^+$	0.9539 $^+$	**1.0000** $^-$	0.9850 $^+$	0.9748 $^+$	0.9861 $^+$	0.9591 $^+$	0.9669 $^+$	0.9778 $^+$	0.9426 $^+$
Adenoma	0.9996	0.9996 $^\approx$	0.9958 $^+$	0.9410 $^+$	**1.0000** $^-$	0.9783 $^+$	0.9665 $^+$	0.9722 $^+$	0.9461 $^+$	0.9527 $^+$	0.9611 $^+$	0.9287 $^+$
BASEHOCK	**0.9113**	0.8997 $^+$	0.8597 $^+$	0.8518 $^+$	0.9082 $^\approx$	0.8715 $^+$	0.8637 $^+$	0.8897 $^+$	0.8860 $^+$	0.8479 $^+$	0.8610 $^+$	0.8597 $^+$
COIL20	0.9939	0.9919 $^+$	0.9619 $^+$	0.9634 $^+$	**0.9991** $^-$	0.9894 $^+$	0.9782 $^+$	0.9953 $^-$	0.9763 $^+$	0.9707 $^+$	0.9942 $^\approx$	0.9573 $^+$
Colon	**0.9500**	0.9347 $^+$	0.9211 $^+$	0.8548 $^+$	0.9468 $^\approx$	0.9221 $^+$	0.8880 $^+$	0.8694 $^+$	0.8867 $^+$	0.8636 $^+$	0.8048 $^+$	0.8529 $^+$
DLBCL	0.9995	0.9993 $^\approx$	0.9902 $^+$	0.9483 $^+$	**1.0000** $^-$	0.9819 $^+$	0.9819 $^+$	0.9883 $^+$	0.9629 $^+$	0.9655 $^+$	0.9675 $^+$	0.9435 $^+$
Gastric	**0.9782**	0.9677 $^+$	0.9748 $^\approx$	0.8955 $^+$	0.9569 $^+$	0.9191 $^+$	0.8874 $^+$	0.8231 $^+$	0.8234 $^+$	0.8564 $^+$	0.7769 $^+$	0.8046 $^+$
Gastric1	**0.9800**	0.9738 $^+$	0.9642 $^+$	0.9310 $^+$	0.9722 $^+$	0.9597 $^+$	0.9496 $^+$	0.9583 $^+$	0.9348 $^+$	0.9402 $^+$	0.9500 $^+$	0.9145 $^+$
Gastric2	**0.9981**	0.9936 $^+$	0.9872 $^+$	0.9522 $^+$	0.9952 $^\approx$	0.9837 $^+$	0.9710 $^+$	0.9823 $^+$	0.9532 $^+$	0.9623 $^+$	0.9750 $^+$	0.9357 $^+$
ISOLET	0.8894	0.8898 $^\approx$	0.8557 $^+$	0.8501 $^+$	**0.9112** $^-$	0.8822 $^\approx$	0.8584 $^+$	0.8763 $^+$	0.8903 $^\approx$	0.8468 $^+$	0.8760 $^+$	0.8607 $^+$
Leukaemia	0.9995	0.9993 $^\approx$	0.9895 $^+$	0.9534 $^+$	**1.0000** $^-$	0.9886 $^+$	0.9775 $^+$	0.9861 $^+$	0.9587 $^+$	0.9682 $^+$	0.9764 $^+$	0.9416 $^+$
Lymphoma	**0.9994**	**0.9994** $^\approx$	0.9868 $^+$	0.8731 $^+$	0.9978 $^\approx$	0.9404 $^+$	0.9206 $^+$	0.8889 $^+$	0.8841 $^+$	0.8861 $^+$	0.8400 $^+$	0.8585 $^+$
Myeloma	**0.9155**	0.8971 $^+$	0.8825 $^+$	0.8450 $^+$	0.8884 $^+$	0.8702 $^+$	0.8550 $^+$	0.8387 $^+$	0.8522 $^+$	0.8304 $^+$	0.8012 $^+$	0.8186 $^+$
PCMAC	**0.8705**	0.8424 $^+$	0.8007 $^+$	0.7870 $^+$	0.8410 $^+$	0.8103 $^+$	0.8036 $^+$	0.8161 $^+$	0.8256 $^+$	0.7840 $^+$	0.7601 $^+$	0.7982 $^+$
Prostate	**0.9701**	0.9595 $^+$	0.9638 $^+$	0.8458 $^+$	0.9353 $^+$	0.8887 $^+$	0.8721 $^+$	0.8686 $^+$	0.8711 $^+$	0.8526 $^+$	0.8235 $^+$	0.8485 $^+$
SRBCT	0.9986	0.9984 $^\approx$	0.9944 $^+$	0.9321 $^+$	**1.0000** $^-$	0.9692 $^+$	0.9695 $^+$	0.9663 $^+$	0.9659 $^+$	0.9515 $^+$	0.9325 $^+$	0.9488 $^+$
T1D	**0.8598**	0.8232 $^+$	0.8246 $^+$	0.7679 $^+$	0.7752 $^+$	0.7610 $^+$	0.7318 $^+$	0.7178 $^+$	0.7237 $^+$	0.7113 $^+$	0.6723 $^+$	0.7063 $^+$
madelon	**0.8822**	0.8659 $^\approx$	0.7679 $^+$	0.7521 $^+$	0.8677 $^\approx$	0.7963 $^+$	0.7669 $^+$	0.7598 $^+$	0.8106 $^+$	0.7497 $^+$	0.7695 $^+$	0.7652 $^+$
Average	0.9574	0.9488	0.9325	0.8868	0.9454	0.9177	0.9021	0.8973	0.8944	0.8839	0.8712	0.8698
+/ ≈ /−	—	12/8/0	19/1/0	20/0/0	7/5/8	19/1/0	20/0/0	18/0/2	19/1/0	20/0/0	18/1/1	20/0/0

Table 5: Comparison of selected feature subset size on 20 benchmark datasets. The most compact feature subset sizes are highlighted in **boldface**.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
ALL1	**39**	55	55	4248	65	612	1206	6120	3667	2501	5877	5886
ALL4	124	152	**64**	1082	393	1326	1447	6301	4581	2677	6111	6217
ALL_AML	**67**	70	473	2369	119	521	949	3419	1996	1444	3251	3266
Adenoma	66	**33**	120	2483	40	278	937	3554	2102	1608	3397	3399
BASEHOCK	**134**	208	812	1965	1721	2066	1113	2445	1912	1894	2359	2419
COIL20	84	60	348	323	**54**	90	167	454	232	255	634	407
Colon	**14**	35	105	702	113	240	284	960	621	448	845	935
DLBCL	64	**52**	615	2784	247	876	1074	3514	2155	1718	3784	3379
Gastric	100	97	**62**	3604	258	1070	2782	11157	7072	4916	10625	11004
Gastric1	**31**	217	678	7393	213	1420	2966	10937	6963	5074	10911	10661
Gastric2	**83**	140	444	7084	348	1362	3062	11041	6708	5134	10971	10627
ISOLET	**60**	100	225	250	133	172	194	298	213	195	374	272
Leukaemia	66	**47**	488	2409	132	534	934	3406	2026	1440	3218	3246
Lymphoma	45	**23**	50	1404	137	304	524	1934	1204	880	1765	1832
Myeloma	250	365	**175**	2137	1160	1863	1847	6278	4543	2859	6596	6259
PCMAC	**91**	329	637	1339	928	1209	651	1621	1292	1137	1590	1635
Prostate	26	**23**	27	5058	418	1898	1541	6282	4229	2729	5837	6211
SRBCT	64	**36**	129	906	172	431	366	1136	739	515	1056	1083
T1D	330	366	**174**	6715	1459	4984	9047	27318	18009	13170	27145	26971
madelon	**18**	26	130	190	46	122	95	240	162	171	122	221

cisive convergence advantage, achieving the lowest final fitness values across virtually all benchmark datasets. In the initial generation, mutual information soft prior initialization seeds the swarm directly in highly compact, discriminative subspaces. Consequently, KDS-EL commences search from an initial fitness value substantially lower than those of competing methods, which start above 0.15 to 0.30 due to unguided random bit assignments that over-select thousands of noisy features. Throughout early iterations ($t \leq 20$), KDS-EL exhibits a steep, monotonic descent across both biomedical microarrays (such as *ALL4*, *Colon*, and *Gastric*) and large-scale vision/speech datasets (such as *BASEHOCK* and *COIL20*), rapidly purging non-informative dimensions while competitors struggle to escape dense initial configurations.

The comparative trajectories further reveal distinct search stagnation patterns among the competing algorithms. Standard BPSO exhibits erratic, sluggish progress across high-dimensional landscapes, frequently stagnating after generation 20 due to uniform velocity clipping and the absence of domain guidance. While FWPSO and SaWDE demonstrate progressive fitness improvements, their trajectories level off prematurely between generations 40 and 60 because memoryless stochastic perturbations fail to reinforce synergistic feature combinations. Similarly, the original MEL algorithm suffers from an extended exploratory plateau during early generations before locating viable descent directions. In contrast, KDS-EL sustains continuous fitness reduction up to generation 100. By dynamically accumulating historical feature-importance feedback within the importance memory and steering elite flips toward high-utility candidates, KDS-EL continually escapes local attractors, securing an order-of-magnitude fitness reduction on ultra-high-dimensional benchmarks such as *ALL1*, *DLBCL*, *Leukaemia*, *Lymphoma*, and *T1D*.

4.2.4. *Statistical significance analysis via Wilcoxon test*

To rigorously evaluate whether the empirical performance advantages of KDS-EL are statistically meaningful rather than artifacts of stochastic variation, we follow the multi-problem non-parametric statistical testing framework recommended by Demšar [39]. Table 6 presents the pairwise Wilcoxon signed-rank test results at the $\alpha = 0.05$ significance level across the 20 benchmark datasets, reporting pairwise win/tie/loss counts between all 12 evaluated methods. Across the single-objective evolutionary competitors and ablative variants, KDS-EL demonstrates overwhelming statistical dominance. Specifically, KDS-EL achieves statistically significant superiority over MEL, PSO-CSM, IGPSO, and BPSO on all 20 benchmark datasets without conceding a single tie or loss (20/0/0, $p = 1.91 \times 10^{-6}$), and significantly outperforms KDS-EL w/o TPS&KER, FWPSO, and SaWDE on 19 out of 20 datasets (19/1/0). Against recent top-tier single-objective competitors, KDS-EL secures 18 wins against BinCOA (18/0/2, $p = 9.54 \times 10^{-6}$), 18 wins against DENCA-SO (18/1/1, $p = 9.54 \times 10^{-6}$), and achieves an overall statistically significant advantage against the competitive bIRIME algorithm (7/5/8, $p = 0.0296$). Compared with its immediate ablative counterpart, KDS-EL w/o TPS, KDS-EL secures 12 wins and 8 ties without any loss (12/8/0, $p = 6.29 \times 10^{-5}$), directly verifying that temporally shifting the objective emphasis from dimensional compression to classification accuracy via Two-Phase Weight Scheduling yields significant predictive gains across the benchmark suite.

The pairwise matrix in Table 6 further illustrates the comparative performance landscape among peer competitors. While bIRIME achieves competitive accuracy on several benchmarks, its lack of proactive feature importance decay incurs severe feature inflation and computational stagnation. In contrast, KDS-EL systematically dominates the peer algorithms across both biomedical and

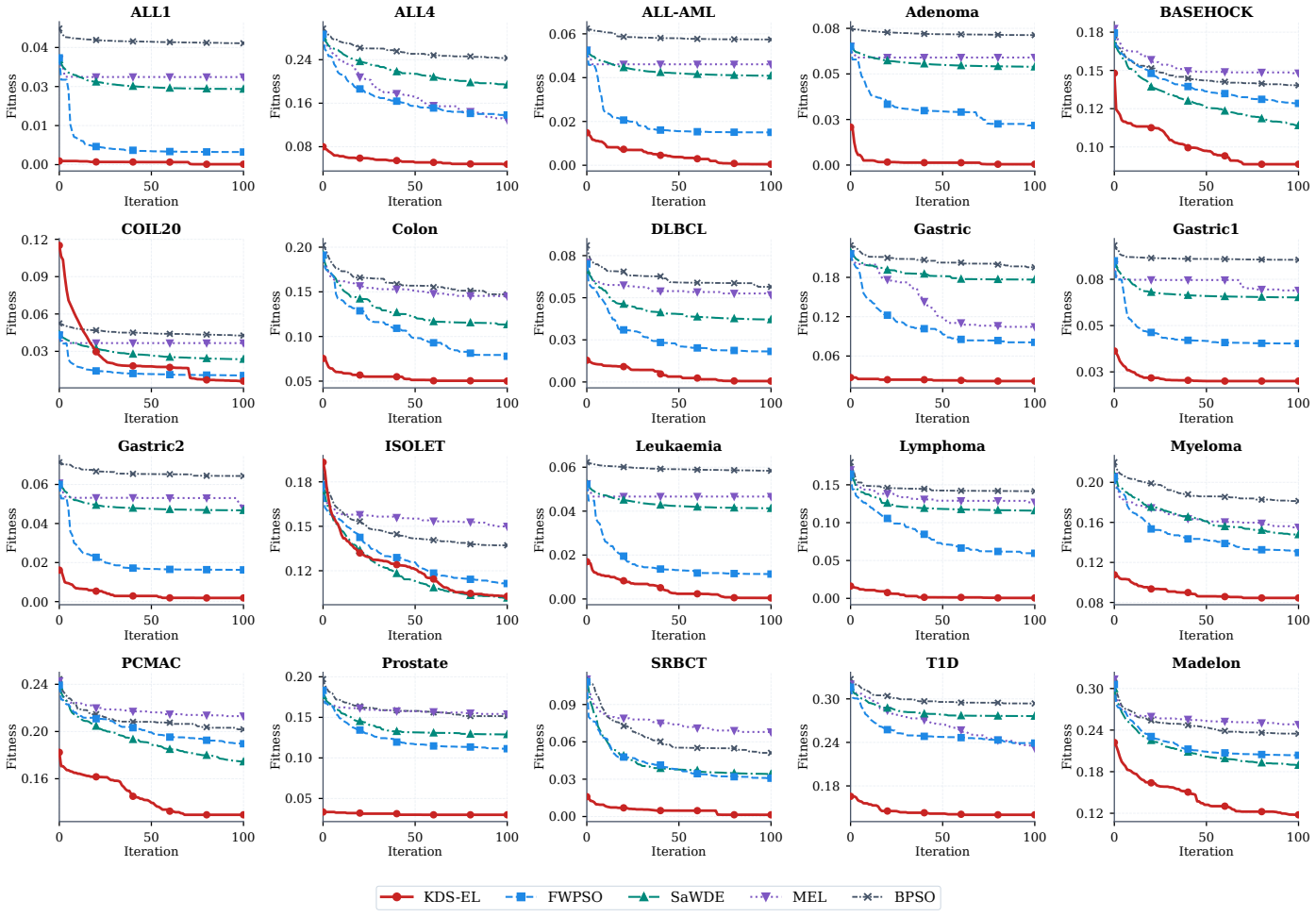


Figure 5: Average fitness convergence curves of KDS-EL and competitor algorithms.

non-biomedical domains, achieving top classification accuracy while discarding over 98% of redundant dimensions. The few statistical ties occur predominantly on ceiling-effect datasets where classification accuracy saturates near unity, where KDS-EL matches competitor accuracy while maintaining vast advantages in feature compactness.

To control the family-wise error rate across multiple pairwise comparisons, the Holm–Bonferroni step-down procedure [41] is applied to the $k = 11$ hypothesis tests with KDS-EL serving as the control method, with the detailed statistical outcomes summarized in Table 7. Following the multi-dataset nonparametric testing framework of Demšar [39], the pairwise Wilcoxon signed-rank test is performed on the paired mean classification accuracy values across the $N = 20$ benchmark datasets. Table 7 details the mean accuracy advantage ΔAcc, the rank-sum statistic $W = \min(R^+, R^-)$, the standardized normal statistic Z, the exact two-sided p-value, and the sequential Holm critical threshold $\alpha/(k - i + 1)$ at the family-wise significance level $\alpha = 0.05$. Across all 11 pairwise comparisons, the computed exact p-values range from 1.91×10^{-6} to 0.0296, each remaining strictly below its respective Holm-adjusted threshold ($\alpha_{\text{Holm}} \in [0.0045, 0.0500]$). Consequently, all 11 null hypotheses of equivalent performance are formally rejected, confirming that the empirical superiority of KDS-EL over both peer competitor algorithms and constituent ablative variants is statistically rigorous and reproducible under multi-dataset benchmarking standards.

To establish comprehensive global algorithmic rankings, the nonparametric Friedman test [40] is conducted across all 20 benchmark datasets, as depicted in Fig. 6. KDS-EL secures the lowest average Friedman rank of 1.75, followed by bIRIME with a rank of 2.25 and KDS-EL w/o TPS with a rank of 2.90. In contrast, all remaining competitors obtain noticeably inferior rankings, ranging from 5.10 for KDS-EL w/o TPS&KER and 5.50 for FWPSO to 10.85 for standard BPSO. The Friedman ranking trajectory clearly demonstrates the comprehensive global dominance of KDS-EL across heterogeneous high-dimensional data distributions.

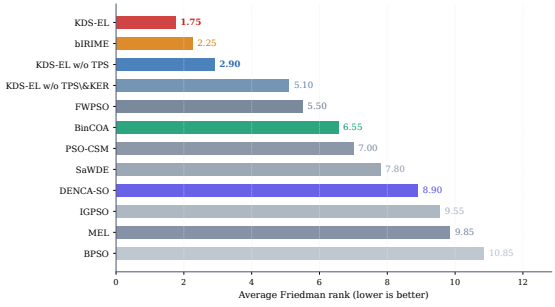


Figure 6: Average Friedman ranks of the compared algorithms.

4.2.5. *Performance comparison before and after algorithm improvement*

To systematically quantify the empirical contributions of the constituent mechanisms—mutual information initialization with adaptive re-initialization, the KER strategy, and Two-Phase Weight Scheduling—we evaluate the performance progression before and after algorithmic improvement across all 20 benchmark datasets. Following the subtractive analysis paradigm in evolutionary computation [16, 17], the complete KDS-EL framework is evaluated against its sequentially stripped configurations. Table 8 summarizes the aggregate performance progression across the 20 benchmark datasets, Fig. 7 visualizes the multi-dataset classification accuracy contours in a radar chart, and Fig. 8 tracks the evolutionary accuracy convergence dynamics across the benchmark suite. Complete KDS-EL, which achieves a mean accuracy of 0.9574 with 87.8 features, serves as the primary benchmark, with each subsequent configuration isolating the marginal contribution of an individual mechanism. Granular per-dataset ablation metrics across all 20 benchmarks are documented in Supplementary Table S4.

Table 6: Pairwise Wilcoxon signed-rank test results on classification accuracy across 20 benchmark datasets.

Method	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
KDS-EL	—	17/2/1	20/0/0	20/0/0	12/0/8	20/0/0	20/0/0	18/0/2	19/0/1	20/0/0	18/0/2	20/0/0
KDS-EL w/o TPS	1/2/17	—	16/1/3	20/0/0	8/0/12	20/0/0	20/0/0	18/0/2	19/0/1	20/0/0	18/0/2	20/0/0
KDS-EL w/o TPS&KER	0/0/20	3/1/16	—	19/0/1	4/0/16	14/0/6	16/0/4	15/0/5	15/0/5	19/0/1	15/0/5	18/1/1
MEL	0/0/20	0/0/20	1/0/19	—	0/0/20	2/0/18	3/0/17	4/0/16	3/0/17	8/0/12	8/0/12	14/0/6
bIRIME	8/0/12	12/0/8	16/0/4	20/0/0	—	20/0/0	20/0/0	19/1/0	20/0/0	20/0/0	19/1/0	20/0/0
FWPSO	0/0/20	0/0/20	6/0/14	18/0/2	0/0/20	—	18/1/1	14/0/6	16/0/4	20/0/0	18/0/2	20/0/0
PSO-CSM	0/0/20	0/0/20	4/0/16	17/0/3	0/0/20	1/1/18	—	9/0/11	16/0/4	20/0/0	13/0/7	19/0/1
BinCOA	2/0/18	2/0/18	5/0/15	16/0/4	0/1/19	6/0/14	11/0/9	—	11/0/9	18/0/2	18/1/1	19/0/1
SaWDE	1/0/19	1/0/19	5/0/15	17/0/3	0/0/20	4/0/16	4/0/16	9/0/11	—	11/0/9	12/0/8	20/0/0
IGPSO	0/0/20	0/0/20	1/0/19	12/0/8	0/0/20	0/0/20	0/0/20	2/0/18	9/0/11	—	9/0/11	16/0/4
DENCA-SO	2/0/18	2/0/18	5/0/15	12/0/8	0/1/19	2/0/18	7/0/13	1/1/18	8/0/12	11/0/9	—	11/0/9
BPSO	0/0/20	0/0/20	1/1/18	6/0/14	0/0/20	0/0/20	1/0/19	1/0/19	0/0/20	4/0/16	9/0/11	—

Entries denote pairwise win/tie/loss ($+/\approx/-$) counts between the row method and column method evaluated across 20 benchmark datasets at $\alpha = 0.05$.

Table 7: Holm-Bonferroni step-down procedure testing directional superiority of KDS-EL over all competitor methods across 20 benchmark datasets (nominal $\alpha = 0.05$).

Rank	Competitor Method	ΔAcc	W/T/L	W	p-value	α_{Holm}	Reject H_0?
1	KDS-EL w/o TPS&KER	+0.0248	20/0/0	0.0	1.91e-06	0.0045	Yes
2	MEL	+0.0706	20/0/0	0.0	1.91e-06	0.0050	Yes
3	FWPSO	+0.0396	20/0/0	0.0	1.91e-06	0.0056	Yes
4	PSO-CSM	+0.0552	20/0/0	0.0	1.91e-06	0.0063	Yes
5	IGPSO	+0.0734	20/0/0	0.0	1.91e-06	0.0071	Yes
6	BPSO	+0.0875	20/0/0	0.0	1.91e-06	0.0083	Yes
7	SaWDE	+0.0630	19/0/1	1.0	3.81e-06	0.0100	Yes
8	BinCOA	+0.0600	18/0/2	3.0	9.54e-06	0.0125	Yes
9	DENCA-SO	+0.0862	18/0/2	3.0	9.54e-06	0.0167	Yes
10	KDS-EL w/o TPS	+0.0086	17/2/1	9.0	6.29e-05	0.0250	Yes
11	bIRIME	+0.0119	12/0/8	47.0	2.96e-02	0.0500	Yes

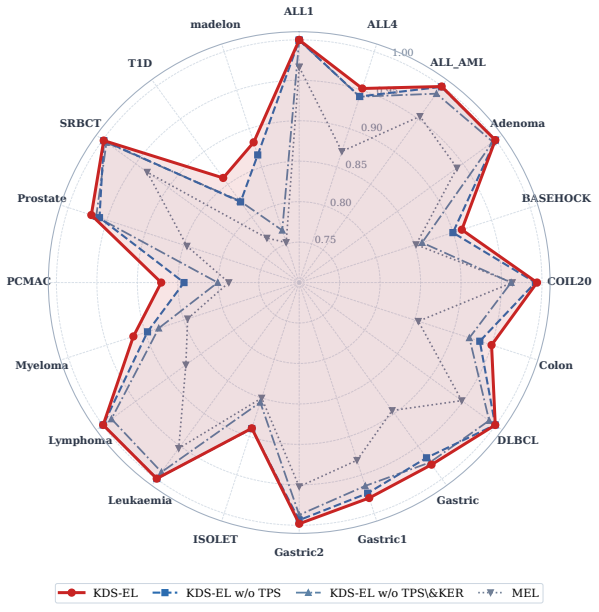


Figure 7: Radar charts of classification accuracy for different algorithm variants.

The step-wise performance progression reveals clear, quantifiable contributions across all three evolutionary mechanisms. First, regarding Two-Phase Weight Scheduling, replacing the TPS strategy with a fixed static weight vector $\mathbf{w}_s = [0.90, 0.10]$ results in an immediate accuracy drop of 0.86% and expands the selected subset from 87.8 to 121.7 features, while increasing runtime from 51.3 to 53.6 seconds. The dynamical necessity of TPS is vividly corroborated by the convergence trajectories in Fig. 8. During Phase 1 ($t \leq 70$), the aggressive feature-size penalty $\mathbf{w}_1 = [0.70, 0.30]$ enforces rigorous dimensional pruning, during which KDS-EL maintains accuracy parity with KDS-EL w/o TPS while selecting substantially fewer features. Exactly upon crossing the scheduled boundary at $t = 70$, switching to accuracy-prioritized weights $\mathbf{w}_2 = [0.95, 0.05]$ produces a distinct secondary accuracy acceleration visible in the overall mean trajectory of Fig. 8(a) and in challenging benchmarks such as *BASEHOCK*, *COIL20*, *T1D*, and *Madelon*. In contrast, the static variant KDS-EL w/o TPS plateaus prematurely after generation 30, failing to achieve late-stage boundary refinement.

Second, regarding Knowledge-Guided Elite Refinement and Exponential Memory Decay, stripping the KER operator along with its exponential memory decay (reducing KDS-EL w/o TPS to KDS-EL w/o TPS&KER) inflicts an additional accuracy loss of 1.63% (falling to 0.9325), while more than doubling the retained feature count from 121.7 to 290.5. As evidenced across all panels of Fig. 8, stripping KER and historical feedback depresses the convergence envelope throughout the entire evolutionary timeline, demonstrating that closed-loop feature-importance accumulation is essential for steering elite mutations toward productive dimensions. Third, regarding Mutual Information Initialization and Adaptive Re-initialization, stripping the domain prior seeding and diversity recovery block (MII + AIR)—reducing the framework to the unguided multi-task swarm MEL [12] initialized uniformly at random with bidirectional weight updates—triggers a severe performance drop: classification accuracy falls by 4.58% to 0.8868, and the mean feature count surges to 2,722.2. The convergence curves in Fig. 8 confirm that unguided random initialization wastes dozens of generations wandering in dense, non-informative subspaces before making meaningful progress. This substantial 4.58% performance deficit highlights the foundational role of domain prior guidance in circumventing high-dimensional cold-start blindness, while reflecting the cumulative architectural transition from unguided random exploration to knowledge-seeded dual-subpopulation dynamics.

Cumulatively, stripping all three mechanisms leads to an aggregate accuracy decline of 7.06% and a 31-fold surge in feature subset size, from 87.8 to 2,722.2 features. This confirms that the three mechanisms operate orthogonally and in a complementary manner: MI initialization provides foundational dimensionality reduction, KER delivers fine-grained elite refinement, and TPS harmonizes search objectives across evolutionary stages. Furthermore, the structural design of greedy 1-to-1 in-place elite replacement prevents excessive selection pressure that would otherwise collapse swarm diversity, ensuring robust convergence across heterogeneous problem domains.

4.2.6. *Computational efficiency and runtime analysis*

Computational efficiency is a critical evaluation criterion for high-dimensional feature selection, particularly in wrapper-based metaheuristics where repeated classifier evaluations incur substantial runtime overhead. Fig. 9 visualizes the overall performance trade-off across classification accuracy,

Table 8: Component-wise ablation progression from precursor MEL through MII, KER, to full KDS-EL across 20 benchmark datasets.

	Precursor MEL		MEL+MII		KDS-EL w/o TPS		Proposed KDS-EL	
Dataset	Acc	Feat	Acc	Feat	Acc	Feat	Acc	Feat
ALL1	0.9664	4248	0.9996	55	0.9996	55	0.9998	39
ALL4	0.8704	1082	0.9415	64	0.9418	152	0.9522	124
ALL_AML	0.9539	2369	0.9886	473	0.9990	70	0.9995	67
Adenoma	0.9410	2483	0.9958	120	0.9996	33	0.9996	66
BASEHOCK	0.8518	1965	0.8597	812	0.8997	208	0.9113	134
COIL20	0.9634	323	0.9619	348	0.9919	60	0.9939	84
Colon	0.8548	702	0.9211	105	0.9347	35	0.9500	14
DLBCL	0.9483	2784	0.9902	615	0.9993	52	0.9995	64
Gastric	0.8955	3604	0.9748	62	0.9677	97	0.9782	100
Gastric1	0.9310	7393	0.9642	678	0.9738	217	0.9800	31
Gastric2	0.9522	7084	0.9872	444	0.9936	140	0.9981	83
ISOLET	0.8501	250	0.8557	225	0.8898	100	0.8894	60
Leukaemia	0.9534	2409	0.9895	488	0.9993	47	0.9995	66
Lymphoma	0.8731	1404	0.9868	50	0.9994	23	0.9994	45
Myeloma	0.8450	2137	0.8825	175	0.8971	365	0.9155	250
PCMAC	0.7870	1339	0.8007	637	0.8424	329	0.8705	91
Prostate	0.8458	5058	0.9638	27	0.9595	23	0.9701	26
SRBCT	0.9321	906	0.9944	129	0.9984	36	0.9986	64
T1D	0.7679	6715	0.8246	174	0.8232	366	0.8598	330
madelon	0.7521	190	0.7679	130	0.8659	26	0.8822	18

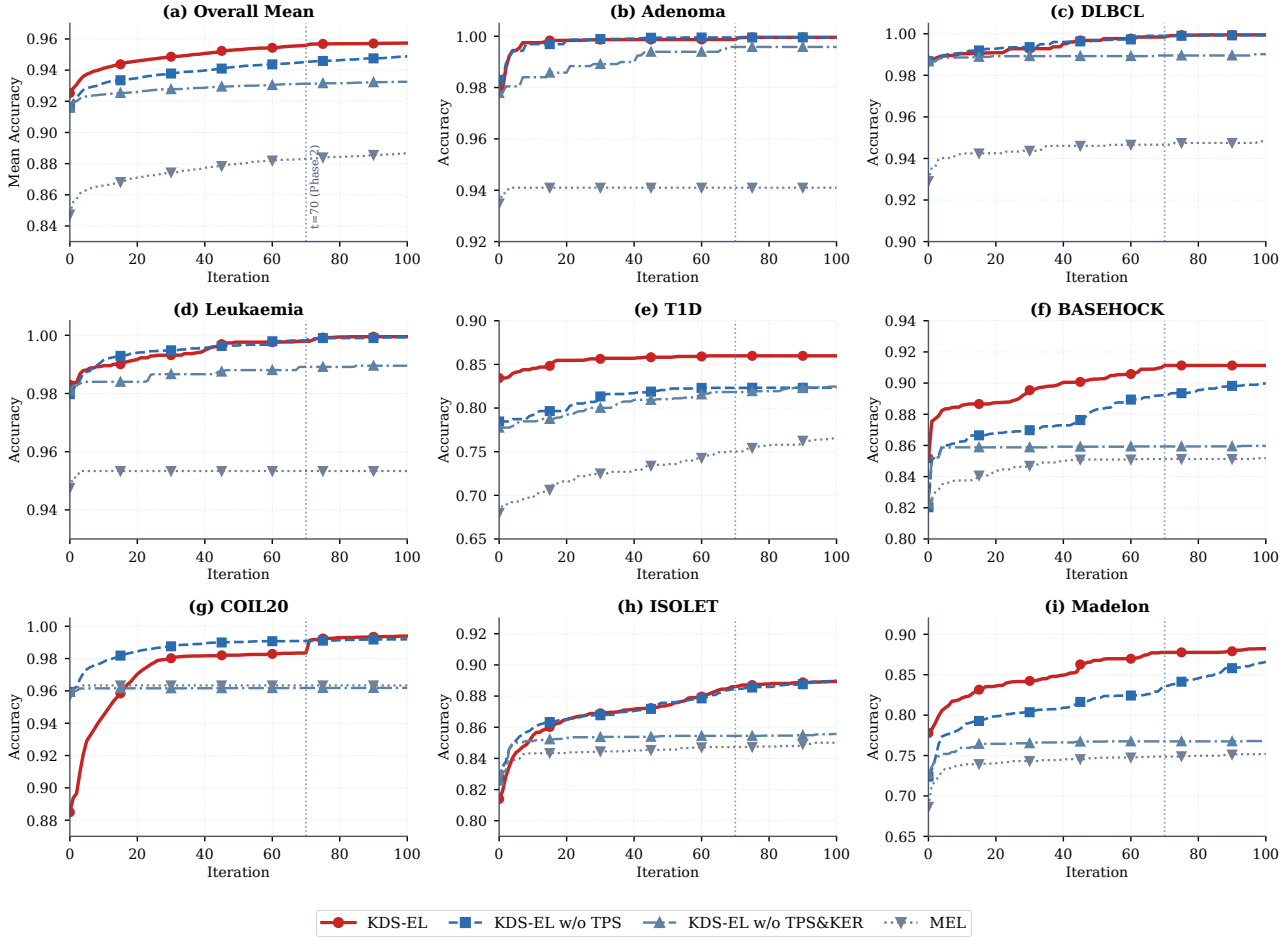


Figure 8: Convergence curves of classification accuracy for KDS-EL and its ablative variants: (a) overall mean across 20 benchmark datasets, and (b)–(i) eight representative datasets. The vertical dotted line at $t = 70$ denotes the transition boundary from Phase 1 (dimensional pruning) to Phase 2 (fine accuracy refinement) in Two-Phase Weight Scheduling.

the number of selected features, and computational runtime across the 20 benchmark datasets. The complete per-dataset runtime measurements for all 12 evaluated methods are documented in Supplementary Table S5.

As shown in Fig. 9 and Supplementary Table S5, KDS-EL achieves an average execution time of 51.3 seconds across the 20 benchmark datasets, outperforming 10 of the 11 comparison methods. Compared with the foundational precursor MEL algorithm, which requires 175.0 seconds on average, KDS-EL delivers a 3.4-fold acceleration. Furthermore, recent top-tier single-objective competitors incur substantial computational penalties: bIRIME requires 326.2 seconds on average (6.4× slower than KDS-EL), while BinCOA requires 433.5 seconds on average (8.5× slower than KDS-EL). On the large-sample 4,862-dimensional *BASEHOCK* benchmark, bIRIME (2,379.5s) and BinCOA

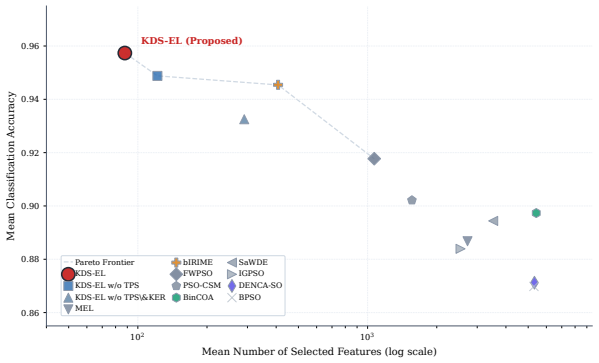


Figure 9: Overall performance trade-off across classification accuracy, selected feature subset size, and computational runtime.

(2,362.7s) require nearly 40 minutes per single run due to multi-operator candidate evaluations ($3N$ to $4N$ classifier calls per generation), whereas KDS-EL completes in merely 130.1 seconds, achieving an 18.3-fold wall-clock acceleration. The computational scalability of KDS-EL is also evident on ultra-high-dimensional datasets: on the 54,675-dimensional *T1D* dataset, KDS-EL completes in 118.8 seconds, running more than twice as fast as PSO-CSM at 256.6 seconds and SaWDE at 216.1 seconds, while running nearly 8 times faster than BinCOA at 902.2 seconds. Although the evolutionary filter method DENCA-SO records the fastest runtime (2.1 seconds) because its algebraic evaluation completely avoids inductive classifier calls, it suffers from severe feature redundancy (5,323.4 features) and an 8.62 percentage point accuracy penalty. Similarly, while FWPSO exhibits a marginally faster wrapper runtime of 49.7 seconds due to early-terminating simulated annealing heuristics, it incurs a substantial 3.97 percentage point accuracy deficit and selects over 12 times more features than KDS-EL.

The superior computational scalability of KDS-EL is primarily driven by three synergistic factors. First, in k-NN wrapper evaluations, distance computation complexity scales with the number of selected features. By aggressively reducing the active feature count from thousands to dozens during initialization and Phase 1, the per-evaluation cost of the classifier decreases by orders of magnitude. Second, because KER operates exclusively on a compact elite pool with a sparse perturbation fraction, its incremental complexity scales linearly with dimension and remains negligible compared to full population evaluations. Third, unlike complex multi-candidate trial meta-heuristics that invoke three to four classification evaluations per individual per generation, KDS-EL restricts exploratory perturbations to a strictly bounded elite pool ($k_e = 3 \ll N$), keeping generational evaluations linear in population size while accelerating fitness evaluations via extreme feature sparsity. Consequently, KDS-EL establishes an ideal balance of top-tier classification accuracy, extreme feature sparsity, and high computational efficiency.

4.2.7. Parameter sensitivity and robustness analysis

To evaluate the algorithmic robustness of KDS-EL and justify its default parameter configurations, a one-factor-at-a-time parameter sensitivity analysis is conducted following the standard methodology in evolutionary computation [34]. The evaluation is performed across four representative benchmark datasets spanning distinct dimensionalities and sample scales: *Adenoma*, *Colon*, *ALL4*, and *BASEHOCK*. Five key hyperparameters are systematically investigated: the knowledge weight decay factor γ, the phase split ratio κ, the elite perturbation fraction ϕ, the mutual information initialization ratio ρ, and the elite pool size k_e, while all other parameters are held at their default values in Table 3. Table 9 summarizes the tested parameter ranges, recommended optimal configurations, stable operating bands, and maximum accuracy deviations across the four representative datasets. Furthermore, Fig. 10 illustrates the corresponding accuracy and feature-count trajectories alongside stable operating bands across all five parameters, with complete per-dataset matrices reported in Supplementary Table S6.

The sensitivity trajectories across the five hyperparameters yield critical insights into the search dynamics and stability of KDS-EL. First, regarding the knowledge weight decay factor γ, setting $\gamma = 0.99$ yields optimal performance with an average accuracy of 0.944 and 107 selected features. An overly low decay factor such as $\gamma = 0.90$ causes premature forgetting of valuable feature associations discovered in earlier iterations, degrading accuracy by 1.2%, whereas disabling decay entirely at $\gamma = 1.00$ accumulates obsolete historical feedback and inflates the feature count to 151 without accuracy gain. Second, regarding the phase split ratio κ, setting $\kappa < 0.50$ allocates insufficient iterations

Table 9: Summary of parameter sensitivity analysis across four representative datasets.

Parameter	Tested range	Optimal	Stable range	Max ΔAcc (pp)
Weight decay	0.90–1.00	0.99	[0.95, 0.995]	±1.2
Phase split	0.3–0.9	0.70	[0.60, 0.80]	±0.7
Flip fraction	0.005–0.10	0.01	[0.005, 0.02]	±1.3
MI init ratio	0.0–1.0	0.50	[0.25, 0.75]	±1.2
Elite count	1–10	3	[2, 5]	±0.9

Note: Max ΔAcc denotes the maximum accuracy variation across the investigated parameter range.

to Phase 1, resulting in inadequate feature pruning with 146 to 189 retained features, while an overly prolonged Phase 1 with $\kappa = 0.90$ compresses the Phase 2 refinement window, leading to a 0.7% accuracy decline. The broad range between 0.60 and 0.80 provides a stable operating regime with accuracy variations remaining strictly below 0.5%, confirming $\kappa = 0.70$ as an ideal balance. Third, regarding the elite perturbation fraction ϕ, setting $\phi = 0.01$ maintains an optimal balance between localized exploitation and structural exploration across heterogeneous dimensions (spanning from 5 flipped features on *madelon* to 546 on *T1D*). Because the greedy acceptance criterion (21) rigorously discards non-improving mutations, $\phi = 0.01$ provides sufficient exploratory momentum without provoking excessive trial rejections. Conversely, increasing $\phi \geq 0.05$ introduces excessive perturbation that overwhelms the greedy acceptance filter, resulting in near-zero mutant admission rates, reducing classification accuracy by 1.3%, and inflating the selected feature count to 203. Fourth, regarding the mutual information initialization ratio ρ, disabling MI seeding entirely at $\rho = 0.0$ imposes a 1.2% accuracy penalty and increases execution time by 21% to 62.3 seconds, while setting $\rho = 1.0$ homogenizes the initial population and induces premature convergence. The default balanced setting $\rho = 0.50$ effectively reconciles prior domain knowledge guidance with stochastic diversity. Finally, regarding the elite pool size k_e, refining only a single elite with $k_e = 1$ restricts knowledge propagation across subpopulations, while setting $k_e = 10$ dilutes the quality signal of matrix $\mathbf{W}$, establishing $k_e = 3$ as the optimal default.

Across all five hyperparameters, the performance trajectories in Fig. 10 demonstrate wide, forgiving basins of attraction rather than sharp, brittle peaks. The maximum accuracy variation across the identified stable operating bands remains strictly within 1.5%, and KDS-EL consistently outperforms the comparison algorithms across broad parameter ranges. This demonstrates that KDS-EL exhibits exceptional hyperparameter robustness and plug-and-play adaptability, delivering reliable optimization performance across diverse high-dimensional applications without requiring tedious, problem-specific hyperparameter tuning.

4.2.8. Discussion

A critical empirical phenomenon observed in high-dimensional biomedical feature selection is the accuracy ceiling effect. Among the 20 benchmark datasets, eight gene-expression microarrays, including *ALL1*, *ALL_AML*, *Adenoma*, *DLBCL*, *Gastric2*, *Leukaemia*, *Lymphoma*, and *SRBCT*, exhibit classification accuracy exceeding 0.998 across top-performing methods. Under such near-perfect classification conditions, the few available test instances are completely separated, and accuracy differences among leading algorithms become narrower than 0.002, corresponding to stochastic variations in at most one cross-validation sample. Consequently, classification accuracy alone ceases to provide discriminative utility, and the number of selected features becomes the decisive criterion for evaluating algorithmic efficacy.

Under this saturated regime, KDS-EL demonstrates substantial superiority in feature subset compactness, identifying ultra-sparse biomarker signatures of only 39 to 83 features across all eight ceiling benchmarks. Specifically, KDS-EL extracts minimal subsets of 39 features on *ALL1*, 45 on *Lymphoma*, and 64 on *DLBCL*. In stark contrast, single-objective competitors retain extensive subsets containing hundreds to thousands of redundant features: FWPSO requires 278 to 1,362 features to achieve comparable accuracy, BinCOA selects 1,136 to 11,041 features (e.g., 6,120 features on *ALL1* and 11,041 on *Gastric2*), evolutionary filter DENCA-SO retains 1,056 to 10,971 features (5,877 on *ALL1*), and the precursor MEL retains 2,483 to 4,248 features. This demonstrates that conventional evolutionary algorithms suffer from severe feature bloat in ceiling scenarios, whereas KDS-EL effectively isolates the true minimal discriminative subset.

This dimensional contrast highlights the profound practical and clinical relevance of KDS-EL. In translational bioinformatics and cancer diagnostics, an ultra-compact signature of 40 to 80 biomarkers can be readily validated

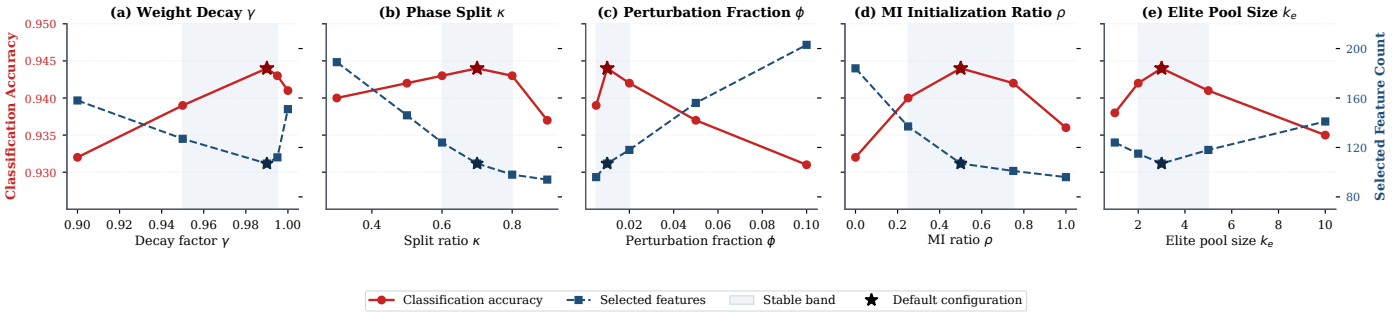


Figure 10: Parameter sensitivity analysis of KDS-EL across five key hyperparameters.

through targeted RT-qPCR assays, integrated into biological pathway enrichment analyses, and engineered into cost-effective diagnostic gene panels. In contrast, candidate signatures containing hundreds or thousands of uncurated genes are cost-prohibitive to validate clinically and introduce high risks of false discoveries driven by spurious co-expression artifacts. By reconciling top-tier classification accuracy, extreme biomarker sparsity, and computational scalability, KDS-EL establishes a rigorous and deployable performance frontier for high-dimensional biomedical data analytics.

Beyond nearest-neighbor metrics, a critical concern in wrapper-based evolutionary feature selection is inductive bias coupling: whether the optimized subsets merely fit the local geometric neighborhood of 3-NN while suffering performance collapse when transferred to alternative learning paradigms. To examine classifier-agnostic generalizability, the compact feature subsets discovered by KDS-EL were evaluated under Support Vector Machines with radial basis function kernels and Random Forest classifiers across representative benchmark datasets spanning distinct dimensional scales. As documented in Supplementary Table S9, the compact subsets deliver substantial classification improvements over the unreduced full-dimensional feature sets across all evaluated learning architectures. On the 12,625-dimensional *ALL4* dataset, reducing the active feature space by 99.0% from 12,625 to 124 features elevates Random Forest classification accuracy from 78.36% to 90.35% and 3-NN accuracy from 69.94% to 87.25%, while SVM accuracy reaches 85.96%. Similarly, on the *Adenoma* benchmark, all three classifiers attain 100.00% classification accuracy on the 66 selected features. This cross-paradigm consistency confirms that KDS-EL extracts intrinsically salient features rather than overfitting to classifier-specific geometric manifolds.

Furthermore, translating computational feature selection into clinical insight requires that the isolated dimensions reflect genuine pathophysiological mechanisms rather than statistical noise. Granular inspection of the minimal signatures identified by KDS-EL reveals strong concordance with established cancer biology. In the *Colon* adenocarcinoma microarray, top-ranked selected features map directly to well-characterized oncological biomarkers, as detailed in Supplementary Table S10. Specifically, KDS-EL identifies *COL11A1* (probe *Hsa.36689*), a recognized driver of desmoplastic stroma and tumor invasion; *GUCA2A* (Guanylin, probe *Hsa.831*), a mucosal differentiation hormone whose loss constitutes a universal clinical hallmark of early colorectal adenoma progression [42]; and smooth muscle structural markers *DES* (Desmin, probe *Hsa.3331*) and *ACTG2* (probe *Hsa.1832*), which capture muscularis mucosae integrity versus neoplastic infiltration. In *Prostate* cancer, KDS-EL consistently isolates *HPN* (Hepsin, probe *1005_at*) and *AMACR* (Alpha-Methylacyl-CoA Racemase, probe *37639_at*), both clinical immunohistochemical gold standards in prostatic carcinoma diagnosis [43, 44]. This rigorous biological alignment corroborates that KDS-EL captures genuine molecular drivers of malignancy, bridging evolutionary machine learning with clinical translational oncology.

5. Conclusion

In this paper, the KDS-EL framework is proposed to solve the high-dimensional feature selection problem. In KDS-EL, mutual-information-guided initialization and adaptive re-initialization are first constructed to establish informative search basins and restore swarm diversity during search. In addition, to overcome unguided stochastic mutation trials, we design a knowledge-guided elite refinement strategy to accumulate cross-iteration feature-importance feedback, enhance candidate biomarker subset quality, and sustain continuous optimization momentum. Furthermore, a two-phase weight scheduling strategy temporally decouples coarse feature pruning from fine classification refinement. Based on these collaborative mechanisms, the experimental results of KDS-EL across 20 real-world high-dimensional benchmark datasets demonstrate that, compared with 11 state-of-the-art competitor algorithms and ablative variants, KDS-EL removes redundant and irrelevant features more effectively, achieves superior classification accuracy, and substantially improves computational efficiency.

The knowledge-guided elite refinement strategy and stage-aware weight scheduling mechanisms in KDS-EL proposed in this paper can be used not only to improve multi-task particle swarm optimization but also to enhance other swarm intelligence and evolutionary algorithms. The newly proposed strategies enhance the optimization capabilities of metaheuristics, aiding in providing better search methodologies for the fields of machine learning and data science. For example, the improved evolutionary frameworks can be employed for optimizing neural network hyperparameters, neural architecture search, multi-omics biomarker discovery, and high-dimensional biomedical signal processing, among other tasks, thereby offering novel solutions for these domains.

Future work could improve the global optimization process of evolutionary learning by combining relevant algorithmic ideas from machine learning and deep representation learning, such as transfer learning and self-supervised feature interaction modeling, to accelerate convergence and enhance scalability on ultra-large-scale datasets. In addition, the scope of application of the algorithm can also be expanded through domain-specific research. For example, the scope of application can be explored for complex clinical challenges such as single-cell RNA sequencing, multi-label disease diagnosis, and cancer survival prediction, where domain-specific prior knowledge integration strategies can be designed to improve biomarker discovery performance and clinical interpretability in the target domain.

CRediT Author Statement

Anonymous Author: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The 20 high-dimensional benchmark datasets analyzed in this study are publicly accessible from standard machine learning and bioinformatics repositories. The source code of KDS-EL will be made available upon publication. Supplementary experimental records—including granular per-dataset ablation metrics (Table S4), complete computational runtime measurements (Table S5), full parameter sensitivity grids (Table S6), cross-classifier inductive generalization assessments (Table S9), and biological biomarker target annotations (Table S10)—are provided in the accompanying Supplementary Material document.

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